UNIVERSIDAD DEL NORTE

MASTER'S RESEARCH PROJECT

Mapping Narrative Patterns in News Media Using BERTopic: A Bibliometric and Topic Modeling Analysis of Scopus Data

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Research project submitted in partial fulfillment of the requirements for the degree in Master of Science in Applied Statistics

at the

Faculty of Basic Sciences
Department of Mathematics and Statistics

June 16, 2026

UNIVERSIDAD DEL NORTE

Abstract

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By Elián David MARTÍNEZ OROZCO

This study addresses the challenge of identifying and analyzing narrative structures in large-scale news media corpora, where the growing volume of digital text data renders manual analysis increasingly impractical. Detecting storytelling patterns in such contexts requires statistical models capable of uncovering latent thematic structures while preserving semantic relationships between documents.

Recent advances in natural language processing have introduced embedding-based approaches that represent textual data in continuous semantic spaces, overcoming several limitations of classical count-based methods. Within this framework, BERTopic provides a modular and flexible pipeline that integrates transformer-based document embeddings, dimensionality reduction, density-based clustering, and class-based term weighting to generate semantically coherent and interpretable topic representations.

This research applies BERTopic to a final corpus of 1,294 Scopus-indexed scientific publications related to storytelling, narratives, and news media. Using abstract-level representations, the model identified 24 substantive topics after excluding the HDBSCAN outlier class. The resulting thematic structure reveals recurrent narrative domains associated with identity and representation, conflict and geopolitics, journalistic trust and authority, public health and risk, environmental issues, and computational approaches to media analysis.

Model quality was assessed through complementary quantitative and qualitative criteria, including topic diversity, topic coherence, and manual inspection of representative terms and abstracts. The results show that BERTopic provides a scalable and statistically grounded framework for examining narrative patterns in large bibliographic corpora while preserving semantic relationships between documents.

This work contributes to computational text analysis and media studies by demonstrating how embedding-based topic modeling can support the systematic study of narrative structures in news media scholarship.

Keywords: BERTopic, Topic Modeling, Narrative Patterns, News Media, Bibliometric Analysis, Scopus, Embedding-Based Methods.

Acknowledgements

I would like to express my sincere gratitude to my thesis advisors, Dr. Humberto Llinás and Dr. Brian Llinás, for their guidance, academic support, and valuable feedback throughout the development of this research. Their observations contributed significantly to the conceptual, methodological, and analytical refinement of this work.

I am also grateful to Universidad del Norte for providing the academic environment and institutional support necessary to carry out this research project.

Special thanks are extended to my classmates and colleagues who shared ideas, suggestions, and discussions during the different stages of this process.

Finally, I would like to thank my family and loved ones for their constant encouragement, patience, and support. Their presence has been essential throughout this academic journey.

Contents

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List of Abbreviations

AI	**A**rtificial **I**ntelligence
BERT	**B**idirectional **E**ncoder **R**epresentations from **T**ransformers
BERTopic	**BERT**-based topic modeling framework
c-TF-IDF	**c**lass-based **T**erm **F**requency-**I**nverse **D**ocument **F**requency
DOI	**D**igital **O**bject **I**dentifier
EM	**E**xpectation-**M**aximization
HDBSCAN	**H**ierarchical **D**ensity-**B**ased **S**patial **C**lustering of **A**pplications with **N**oise
IDF	**I**nverse **D**ocument **F**requency
LDA	**L**atent **D**irichlet **A**llocation
LLM	**L**arge **L**anguage **M**odel
LSA	**L**atent **S**emantic **A**nalysis
MCA	**M**ultiple **C**orrespondence **A**nalysis
MCP	**M**ultiple **C**ountry **P**ublications
NER	**N**amed **E**ntity **R**ecognition
NLP	**N**atural **L**anguage **P**rocessing
NPMI	**N**ormalized **P**ointwise **M**utual **I**nformation
PCA	**P**rincipal **C**omponent **A**nalysis
pLSA	**p**robabilistic **L**atent **S**emantic **A**nalysis
PRISMA	**P**referred **R**eporting **I**tems for **S**ystematic Reviews and **M**eta-**A**nalyses
SBERT	**S**entence-**BERT**
SCP	**S**ingle **C**ountry **P**ublications
SVD	**S**ingular **V**alue **D**ecomposition
TF	**T**erm **F**requency
TF-IDF	**T**erm **F**requency-**I**nverse **D**ocument **F**requency
t-SNE	**t**-distributed **S**tochastic **N**eighbor **E**mbedding
UMAP	**U**niform **M**anifold **A**pproximation and **P**rojection
VOSviewer	**V**isualization **o**f **S**imilarities viewer

To my family, for their love, patience, and unwavering support.

…

Chapter 1

Introduction

The rapid growth of digital text data has transformed the way information is produced, disseminated, and consumed, particularly in the context of news media. Contemporary journalism generates vast amounts of textual content that reflect complex narrative structures, temporal dynamics, and socio-political perspectives. Understanding these narrative patterns is essential for analyzing how news shapes public discourse and constructs meaning across different contexts. However, the scale and heterogeneity of modern textual data pose significant challenges for traditional analytical approaches.

From a computational perspective, the problem addressed in this study is the identification and modeling of latent narrative structures within large collections of news-related texts. Specifically, the objective is to uncover thematic patterns that reflect storytelling dynamics while preserving the semantic relationships between documents. Early approaches to text representation, grounded in the distributional hypothesis (Harris, 1954), and the vector space model (Salton, Wong, & Yang, 1975), laid the foundation for statistical analysis of textual data. These ideas evolved into latent semantic methods such as Latent Semantic Analysis (Deerwester, Dumais, Furnas, Landauer, & Harshman, 1990) and probabilistic topic models, particularly Latent Dirichlet Allocation (LDA) (Blei et al., 2003), which have been widely used for topic discovery.

Despite their relevance, these classical models rely on simplifying assumptions, such as the bag-of-words representation and conditional independence between words, which limit their ability to capture contextual meaning in complex narrative texts. Extensions such as Dynamic Topic Models (Blei & Lafferty, 2006) attempt to incorporate temporal dynamics, yet they still struggle to preserve semantic richness and contextual coherence. Consequently, there exists a clear gap in the literature: the need for models that can identify latent thematic structures while maintaining semantic relationships, particularly in narrative-driven domains such as news media.

Recent advances in natural language processing have addressed these limitations through distributed representations of language. Word embeddings (Mikolov, Sutskever, Chen, Corrado, & Dean, 2013; Pennington, Socher, & Manning, 2014) enable the representation of words in continuous vector spaces, capturing semantic similarity beyond simple co-occurrence patterns. These developments were further enhanced by transformer-based architectures (Vaswani et al., 2017), which allow for contextualized representations of text. Models such as BERT (Devlin et al., 2019) generate deep bidirectional embeddings that capture meaning at both the sentence and document levels, providing a richer representation of textual data.

Within this context, BERTopic (Grootendorst, 2022) emerges as a flexible and modular framework for topic modeling. It combines transformer-based embeddings with dimensionality reduction and density-based clustering methods such as HDBSCAN (Campello, Moulavi, & Sander, 2013), along with class-based TF-IDF weighting, to generate interpretable and semantically coherent topics. Unlike classical approaches, BERTopic leverages contextual embeddings, allowing it to better capture the complexity of narrative structures in large and heterogeneous corpora.

The relevance of this study lies in its contribution to the analysis of storytelling patterns in news media using modern embedding-based approaches. News narratives are inherently complex and multidimensional, shaped by temporal, cultural, and ideological factors. Traditional models often fail to fully capture these dimensions, whereas recent research highlights the effectiveness of embedding-based and transformer-driven methods in preserving semantic relationships and improving topic coherence in news-related data.

Based on this context, the general objective of this research is to model and analyze narrative structures in scientific publications related to news media using BERTopic, with the aim of identifying latent thematic patterns that reflect narrative structures within the corpus. The study is guided by the following research question:

- What latent narrative patterns can be identified in scientific publications related to news media using BERTopic, and how do these topics reflect underlying narrative structures?

The choice of BERTopic is justified by its ability to overcome the limitations of classical topic models, particularly in terms of semantic representation and interpretability. Its modular architecture allows the integration of state-of-the-art embedding models and clustering techniques, providing a robust framework for narrative analysis.

Finally, the empirical analysis is based on data retrieved from the Scopus database, one of the most comprehensive and curated repositories of scientific publications (Mongeon & Paul-Hus, 2016). The use of Scopus ensures data quality, interdisciplinary coverage, and relevance, which are essential for conducting a reliable large-scale analysis of narrative structures in scientific literature related to news media.

This work is structured as follows. First, a corpus of scientific publications related to news media and narrative analysis is retrieved from the Scopus database. The textual data is then preprocessed through standard techniques, including lowercasing, removal of punctuation and stopwords, and normalization of whitespace. Subsequently, documents are transformed into contextual embeddings using a transformer-based model, specifically Sentence-BERT (`all-MiniLM-L6-v2`), allowing for the representation of semantic relationships within the corpus. These high-dimensional embeddings are projected into a lower-dimensional space using Uniform Manifold Approximation and Projection (UMAP), facilitating the identification of structure in the data. Next, density-based clustering is performed using HDBSCAN to group semantically similar documents into latent topics. For each identified cluster, topic representations are constructed through a class-based TF-IDF weighting scheme, enabling the extraction of the most representative terms. Finally, the resulting topics are analyzed and visualized to interpret the underlying narrative structures, and their quality is assessed using coherence and diversity measures. A detailed description of each methodological stage is presented in the Chapter 3.

Chapter 2

Statistical, Mathematical, and Theoretical Foundations

This chapter presents the statistical and theoretical foundations underlying the methodology of this study. It begins by tracing the evolution of topic modeling, from early frequency-based representations of text to probabilistic generative models, establishing the conceptual foundations upon which modern approaches are built. Building on this progression, the chapter introduces BERTopic as the central analytical framework, detailing the mathematical foundations of its main components: transformer-based document embeddings, dimensionality reduction via UMAP, density-based clustering via HDBSCAN, and class-based term weighting through c-TF-IDF. Together, these sections provide the statistical and geometric intuition necessary to understand the modeling decisions and to interpret the results presented in subsequent chapters.

2.1 The Statistical Evolution of Topic Modeling

The analysis of textual data has long occupied a central place in statistics, information science, and computational linguistics. Long before the emergence of contemporary language models, researchers were already concerned with a fundamental question: how can language be transformed into a form that allows systematic and quantitative analysis? Early work in information retrieval provided one of the first systematic answers by proposing that documents could be represented as numerical objects. In particular, the vector space model introduced by Salton et al. (1975) described documents as vectors in a high-dimensional space, where each dimension corresponds to a term and similarity between documents can be measured geometrically. This representation marked an important conceptual shift, moving language analysis away from purely symbolic representations toward mathematical modeling.

Within this emerging framework, the importance of term weighting quickly became apparent. Not all words contribute equally to the meaning of a document, and treating them as such leads to poor representations. Sparck Jones (1972) addressed this problem by introducing inverse document frequency, a statistical measure designed to capture how informative a term is across a collection of documents. This idea later became a cornerstone of term weighting schemes and was further developed in classical information retrieval texts, such as the work of Salton and McGill (1983), where document-term matrices, similarity measures, and ranking models were formalized in a coherent analytical framework. Although these approaches were not probabilistic in nature, they established a practical and mathematically grounded way of representing text.

At the same time, a probabilistic view of language was gaining traction. Rather than treating documents as static vectors, researchers began modeling text as the outcome of random processes. Early statistical language models described word sequences using multinomial distributions and conditional probabilities, an approach exemplified by the class-based n-gram models developed by Brown, deSouza, Mercer, Della Pietra, and Lai (1992). This probabilistic perspective was later systematized in foundational works such as that of Manning and Schütze (1999), which emphasized estimation, smoothing, and independence assumptions as central elements of statistical natural language processing. These ideas introduced a new way of thinking about language, not as a fixed structure, but as data generated by underlying stochastic mechanisms.

A further conceptual advance came with the introduction of Latent Semantic Analysis. Deerwester et al. (1990) proposed that the observed patterns of word usage in documents are driven by a lower-dimensional latent structure. By applying singular value decomposition to the term-document matrix, LSA revealed semantic relationships not immediately visible in raw frequency counts. This approach demonstrated that meaning could be uncovered through dimensionality reduction and linear algebra, offering a powerful alternative to wholly frequency-based representations. However, despite its success, LSA lacked a probabilistic interpretation and did not provide a principled way to quantify uncertainty.

The desire to combine latent semantic structure with probabilistic modeling led to the development of Probabilistic Latent Semantic Analysis. Hofmann (1999) reformulated latent semantic modeling as a mixture model, in which documents are expressed as mixtures of latent topics and words are generated conditionally on those topics. This formulation introduced latent variables and likelihood-based inference into topic modeling, typically estimated using the Expectation-Maximization algorithm. While pLSA addressed several limitations of LSA, it still lacked a fully generative framework, particularly in modeling document-level variability.

Underlying all these developments is a broader statistical tradition concerned with inference, uncertainty, and latent structure. Foundational contributions in statistical pattern recognition and Bayesian learning, such as those by Bishop (1995) and MacKay (2003), provided the theoretical foundations necessary to formalize hierarchical models, prior distributions, and information-theoretic interpretations of learning algorithms. These ideas created the conditions for the emergence of fully Bayesian approaches to text modeling.

Taken together, the developments preceding Latent Dirichlet Allocation show that modern topic modeling [1] emerged from a gradual convergence of ideas from information retrieval, probabilistic language modeling, linear algebra, and Bayesian inference. Understanding this lineage is essential for appreciating later advances in topic modeling and for situating contemporary approaches within a coherent statistical framework. This foundation also provides the necessary context for examining how subsequent models move beyond classical generative assumptions toward representation-driven methods grounded in geometric and embedding-based representations of language.

[1] For readers interested in extending this historical line, relevant references include early distributional and information retrieval foundations (Harris, 1954; Salton et al., 1975; Sparck Jones, 1972), latent semantic and probabilistic topic models (Blei, 2012; Blei & Lafferty, 2006; Blei et al., 2003; Deerwester et al., 1990; Griffiths & Steyvers, 2004; Hofmann, 1999), and embedding-based or neural approaches to text representation (Devlin et al., 2019; Grootendorst, 2022; Mikolov, Chen, Corrado, & Dean, 2013; Pennington et al., 2014; Reimers & Gurevych, 2019; Vaswani et al., 2017).

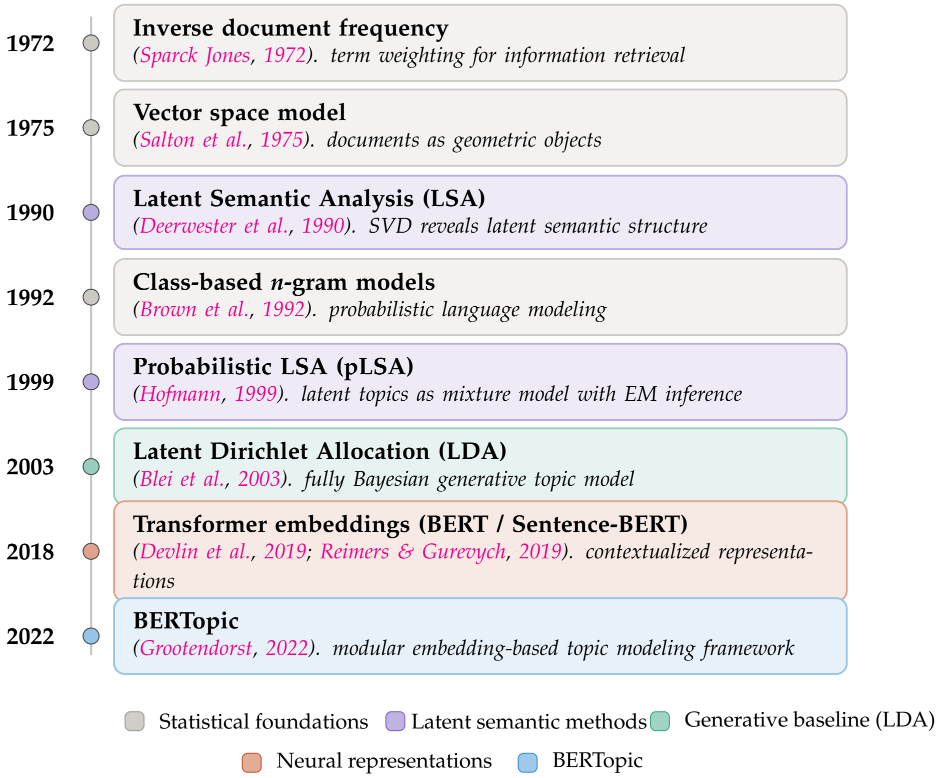


FIGURE 2.1: From statistical foundations to embedding-based approaches. Source: Authors' own elaboration.

2.1.1 Latent Dirichlet Allocation: The Generative Baseline

Introduced by Blei et al. (2003), LDA is a generative probabilistic model that represents a corpus of M documents over a vocabulary of V terms as a mixture of K latent topics. The following notation is used throughout this section. Each topic $k \in \{1, \dots, K\}$ is represented by a word-probability vector defined as

$$\boldsymbol{\phi}_k = (\phi_{k1}, \dots, \phi_{kV}), \qquad \phi_{kv} = P(w = v \mid z = k). \tag{2.1}$$

Here, ϕ_{kv} denotes the probability of term v under topic k, and $\sum_{v=1}^{V} \phi_{kv} = 1$. Each document $d \in \{1, \dots, M\}$ is characterized by a topic proportion vector $\boldsymbol{\theta}_d = (\theta_{d1}, \dots, \theta_{dK})$, where $\theta_{dk} = P(z = k \mid d)$ denotes the proportion of topic k in document d, and $\sum_{k=1}^{K} \theta_{dk} = 1$.

Both $\boldsymbol{\theta}_d$ and $\boldsymbol{\phi}_k$ are modeled as random variables drawn from Dirichlet distributions. In this context, a Dirichlet distribution is used because it defines a probability distribution over vectors whose components are non-negative and sum to one (Blei et al., 2003; Minka, 2000):

$$\boldsymbol{\theta}_d \sim \text{Dirichlet}(\boldsymbol{\alpha}), \qquad \boldsymbol{\phi}_k \sim \text{Dirichlet}(\boldsymbol{\beta}), \tag{2.2}$$

where $\boldsymbol{\alpha} \in \mathbb{R}_+^K$ and $\boldsymbol{\beta} \in \mathbb{R}_+^V$ are concentration hyperparameters that control, respectively, the sparsity of the topic mixtures within documents and the sparsity of the word distributions within topics.

Under this generative framework, each word $w_{d,n}$ in document d is generated through a latent topic assignment. For the n-th word, LDA first draws a topic indicator $z_{d,n}$ from the document-specific topic distribution,

$$z_{d,n} \mid \boldsymbol{\theta}_d \sim \text{Categorical}(\boldsymbol{\theta}_d).^{2} \tag{2.3}$$

In this context, a categorical distribution is a probability model for selecting one outcome from a finite set of categories (Bishop, 2006; Murphy, 2012). In LDA, the possible categories are the K latent topics (Blei et al., 2003). For example, if a document has topic proportions $\boldsymbol{\theta}_d = (0.6, 0.3, 0.1)$ for three topics, then each word position has a 60% probability of being assigned to Topic 1, a 30% probability of being assigned to Topic 2, and a 10% probability of being assigned to Topic 3. Thus, $\boldsymbol{\theta}_d$ determines how likely each latent topic is to be selected for a word in document d.

Once the topic assignment has been selected, the observed word is then drawn from the word distribution associated with that topic,

$$w_{d,n} \mid z_{d,n}, \boldsymbol{\phi} \sim \text{Categorical}(\boldsymbol{\phi}_{z_{d,n}}).^{3} \tag{2.4}$$

In this way, each document is represented as a mixture of topics, and each topic is represented as a probability distribution over words.

The joint distribution over the topic-word distributions, document-topic distributions, latent topic assignments, and observed words is given by

$$p(\boldsymbol{\phi}_{1:K}, \boldsymbol{\theta}_{1:M}, \mathbf{z}_{1:M}, \mathbf{w}_{1:M}) = \prod_{k=1}^{K} p(\boldsymbol{\phi}_k) \prod_{d=1}^{M} p(\boldsymbol{\theta}_d) \prod_{n=1}^{N_d} p(z_{d,n} \mid \boldsymbol{\theta}_d)\, p(w_{d,n} \mid \boldsymbol{\phi}_{1:K}, z_{d,n}), \tag{2.5}$$

where $\mathbf{z}_{1:M}$ denotes the collection of latent topic assignments and $\mathbf{w}_{1:M}$ denotes the observed words across the corpus. Figure 2.2 summarizes this generative structure using plate notation, making explicit the conditional independence assumption of LDA. In this context, conditional independence means that, once the topic assignment $z_{d,n}$ and the corresponding topic-specific word distribution $\boldsymbol{\phi}_{z_{d,n}}$ are known, the probability of the word $w_{d,n}$ does not depend directly on the other words in the document. This reflects the Bag-of-Words (BoW) structure of LDA, where word order and direct dependencies between words are not explicitly modeled.

Although Latent Dirichlet Allocation provides a principled probabilistic framework for uncovering latent thematic structure, several of its limitations stem directly from the assumptions that make the model tractable. First, LDA relies on a bag-of-words representation, in which documents are treated as unordered collections of tokens and word order is ignored. This exchangeability assumption simplifies inference but removes syntactic and sequential information that is often essential for interpreting meaning in natural language (Blei, 2012; Blei et al., 2003).

Second, semantic relationships between words are captured only indirectly through co-occurrence statistics. Consequently, words that are semantically related but rarely appear together may be separated across topics, while frequent but weakly informative co-occurrences can dominate topic formation. This constraint

[2] The notation $z_{d,n} \mid \boldsymbol{\theta}_d$ is read as "$z_{d,n}$ given $\boldsymbol{\theta}_d$". It indicates that the topic assignment for the n-th word in document d is drawn conditional on the document-specific topic proportions $\boldsymbol{\theta}_d$.

[3] The notation $w_{d,n} \mid z_{d,n}, \boldsymbol{\phi}$ is read as "$w_{d,n}$ given $z_{d,n}$ and $\boldsymbol{\phi}$". It means that once the latent topic of the word is known, the actual word is selected from the vocabulary according to the probability distribution of that topic.

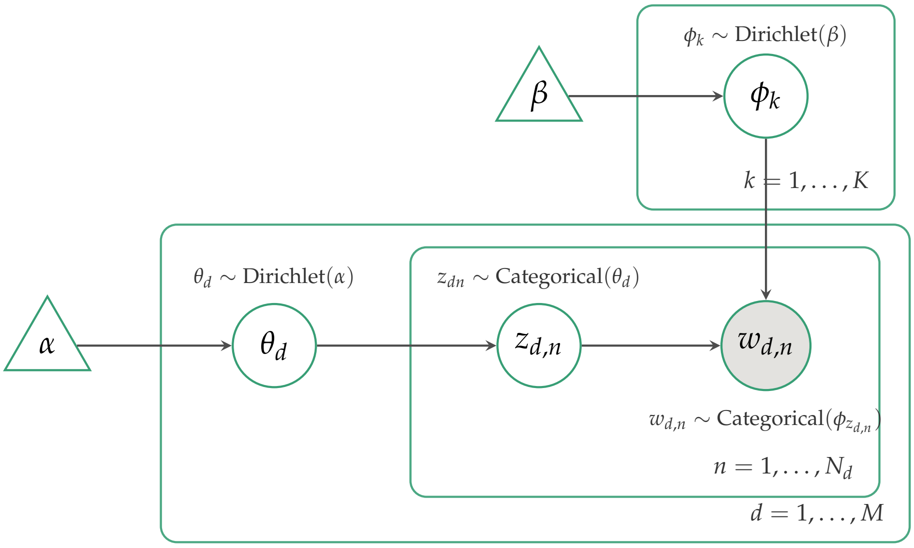


FIGURE 2.2: Plate notation for Latent Dirichlet Allocation. The shaded node $w_{d,n}$ denotes the observed variable; all remaining nodes are latent. Plates represent replication: the inner plate replicates over $n = 1, \ldots, N_d$ words per document, the outer plate replicates over $d = 1, \ldots, M$ documents, and the upper plate replicates over $k = 1, \ldots, K$ topics. Adapted from Blei et al. (2003). Source: Authors' own elaboration.

makes it difficult for LDA to account for phenomena such as synonymy and polysemy, particularly in large or heterogeneous corpora.

Finally, LDA embeds both topics and documents within a probability simplex. While this representation is mathematically well-defined, distances within the simplex do not necessarily correspond to semantic similarity, as topics that are close in a probabilistic sense may still be semantically distinct. Moreover, because the model operates in a high-dimensional and sparse count space, its performance is sensitive to preprocessing decisions such as vocabulary selection, stopword removal, and document length normalization. Taken together, these limitations motivate a paradigm shift from generative count-based models toward approaches grounded in continuous semantic representations.

2.2 BERT and Contextualized Language Representations

The limitations of static word embeddings, most notably their inability to represent polysemy and contextual variation, were addressed by the introduction of transformer-based language models. The transformer architecture (Vaswani et al., 2017) replaces recurrent and convolutional processing with a self-attention mechanism that computes weighted relationships between all positions in a sequence simultaneously. For a sequence of token representations, the scaled dot-product attention output is defined as:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \tag{2.6}$$

where Q, K, and V denote the query, key, and value matrices derived from the input, and d_k is the dimensionality of the key vectors. For a sequence of length n, both Q and K have dimensions $n \times d_k$, so the product $QK^\top$ yields an $n \times n$ matrix containing pairwise attention scores between all tokens in the sequence. The scaling factor $\sqrt{d_k}$ normalizes these scores, preventing them from growing excessively as the embedding dimension increases and thereby improving the stability of the subsequent softmax operation during training.

This formulation allows each token to integrate information from every other token in the sequence, weighted by learned relevance scores, enabling the model to capture long-range dependencies and context-sensitive meaning that earlier architectures could not represent.

Building on this architecture, Devlin et al. (2019) introduced BERT (Bidirectional Encoder Representations from Transformers), a model pre-trained on large text corpora using two self-supervised objectives: masked language modelling, in which randomly selected tokens are predicted from their surrounding context, and next-sentence prediction. The key contribution of BERT lies in its bidirectionality: unlike earlier language models that processed text in a single direction, BERT conditions each token representation on both its left and right context simultaneously, yielding richer and more semantically coherent representations. Figure 2.3 illustrates the overall architecture, in which input tokens are first mapped to embeddings E_i and then processed by a stack of bidirectional transformer encoder layers, each producing a contextualized output representation T_i. The special `[CLS]` token aggregates sequence-level information and is typically used for sentence-level classification tasks, while individual token representations $T_1, \ldots, T_N$ support token-level tasks such as named-entity recognition and question answering.

However, BERT was not designed to produce meaningful fixed-length representations at the sentence or document level. Reimers and Gurevych (2019) addressed this limitation by introducing Sentence-BERT (SBERT), which fine-tunes the BERT encoder using a Siamese network structure on natural language inference tasks. The resulting model produces sentence-level embeddings in which geometric proximity reflects semantic similarity, a property that is essential for the clustering-based architecture of BERTopic and that motivates its adoption as the first stage of the pipeline described in the following section.

2.3 BERTopic: A Modular Framework for Semantic Topic Discovery

The transition from probabilistic generative models to embedding-based approaches represents a fundamental shift in how textual data is represented and analyzed. Classical models such as LDA rely on discrete count-based representations, encoding documents as sparse vectors over a fixed vocabulary. This representation, while tractable, does not model the contextual and sequential structure of language that is essential for capturing narrative meaning. Distributed representations offer an alternative grounding: by mapping text into continuous vector spaces, semantic similarity can be interpreted as a geometric property, enabling richer and more expressive models of thematic structure.

Word embeddings such as Word2Vec (Mikolov, Sutskever, et al., 2013) and GloVe (Pennington et al., 2014) demonstrated that semantic relationships can be encoded as spatial proximity in a continuous vector space. However, these representations

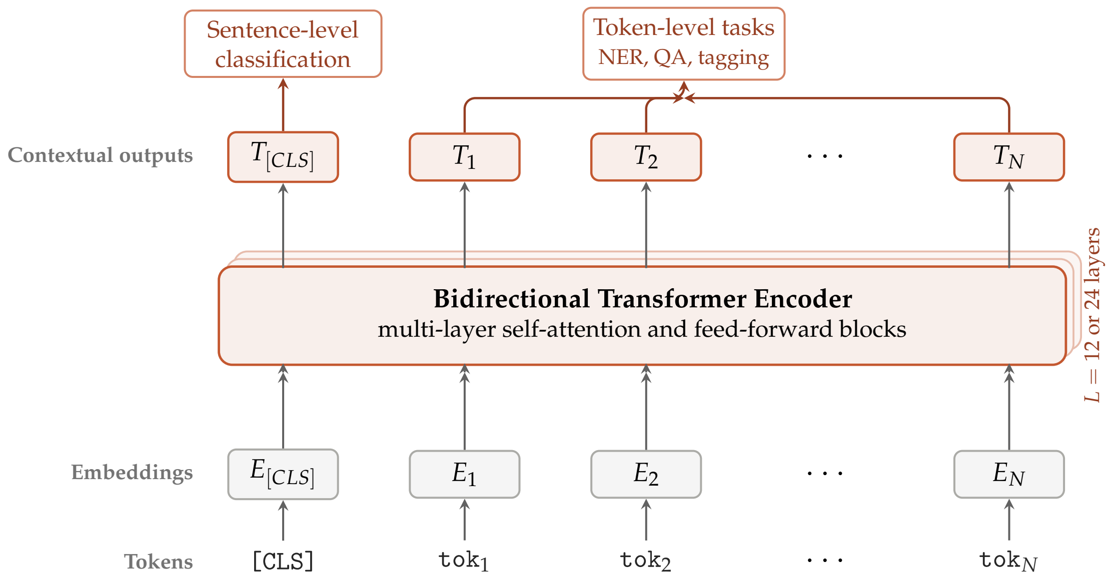


FIGURE 2.3: Architecture of BERT. Adapted from Devlin et al. (2019).
Source: Authors' own elaboration.

are static, as the same word receives the same vector regardless of context, limiting their ability to capture polysemy and contextual variation. Transformer-based architectures (Vaswani et al., 2017) resolved this limitation by computing context-dependent representations through multi-head self-attention, and BERT (Devlin et al., 2019) extended this framework to produce contextualized representations at the sentence and document level through deep bidirectional pre-training. Sentence-BERT (Reimers & Gurevych, 2019) further refined this approach by optimizing document-level embeddings for semantic similarity tasks, providing the embedding space that serves as the representational foundation of BERTopic upon which BERTopic is built.

BERTopic (Grootendorst, 2022) defines a modular topic modeling pipeline that decouples the modeling process into four conceptually distinct stages: document embedding, dimensionality reduction, density-based clustering, and topic representation and labeling. Each stage is statistically grounded and independently interchangeable, allowing the framework to be adapted to diverse corpora and analytical objectives without modifying the underlying architecture. The following subsections develop the mathematical and geometric foundations of each component, tracing how raw text is progressively transformed into semantically coherent and interpretable topic representations. Throughout this section, notation is introduced progressively and used consistently to support clarity and interpretability across the mathematical formulation of the pipeline.

2.3.1 Stage 1: Document Embeddings via Sentence-BERT

The first stage of the pipeline maps each document into a high-dimensional continuous semantic space. BERTopic relies on Sentence-BERT (SBERT) (Reimers & Gurevych, 2019), which extends the BERT encoder through a Siamese network architecture trained on natural language inference tasks to produce semantically meaningful fixed-length representations at the sentence or document level.

Let d_i denote the i-th document in a corpus of size N. SBERT maps each document d_i to a dense vector representation $\mathbf{x}_i$ defined as:

$$\mathbf{x}_i = f(d_i) \in \mathbb{R}^H \tag{2.7}$$

where $f(\cdot)$ denotes the SBERT encoding function and H is the embedding dimension (typically $H = 384$). The vector $\mathbf{x}_i \in \mathbb{R}^H$ represents the semantic embedding of document d_i in a continuous vector space.

The critical property of this representation is that geometric proximity in $\mathbb{R}^H$ reflects semantic similarity: documents with related content are mapped to nearby regions of the space, regardless of lexical overlap. This distinguishes embedding-based models from count-based representations, where similarity is defined over shared vocabulary rather than shared meaning.

It is important to emphasize that all notation introduced in this section is explicitly defined and consistently used throughout the chapter.

2.3.2 Stage 2: Dimensionality Reduction via UMAP

High-dimensional embeddings suffer from the curse of dimensionality: as H increases, Euclidean distances tend to concentrate, becoming increasingly uniform across data points and degrading the performance of subsequent clustering algorithms. To address this, BERTopic applies Uniform Manifold Approximation and Projection (UMAP) (McInnes, Healy, & Melville, 2018), which maps each high-dimensional embedding $\mathbf{x}_i \in \mathbb{R}^H$ into a reduced representation $\mathbf{z}_i \in \mathbb{R}^d$, whereas preserving the local structure of the data.

UMAP proceeds in two main phases. In the first phase,it constructs a weighted fuzzy simplicial set over the data, assigning a membership strength $v_{ij} \in [0, 1]$ to each pair of documents (i, j), based on their distance relative to the k-nearest neighbor structure. This representation encodes the local connectivity of the high-dimensional space. This membership strength reflects the probability that documents i and j are considered neighbors in the high-dimensional space. In the second phase, UMAP seeks a low-dimensional embedding $\{\mathbf{z}_i\}_{i=1}^N$ by minimizing the cross-entropy between the between the high-dimensional fuzzy representation and its low-dimensional counterpart:

$$\mathcal{L}_{\text{UMAP}} = \sum_{i=1}^{N} \sum_{j=1}^{N} \left[v_{ij} \log \frac{v_{ij}}{w_{ij}} + (1 - v_{ij}) \log \frac{1 - v_{ij}}{1 - w_{ij}} \right], \tag{2.8}$$

where $v_{ij} \in [0, 1]$ denotes the membership strength of the pair (i, j) in the high-dimensional space, and $w_{ij} \in [0, 1]$ denotes the corresponding membership strength induced by low-dimensional embedding $\{\mathbf{z}_i\}$.

This objective simultaneously preserves local structure, ensuring that close neighbor in $\mathbb{R}^H$ remain close in $\mathbb{R}^d$. This objective emphasizes the preservation of local structure, while still maintaining a degree of global organization. This represents an advantage over Principal Component Analysis (PCA), which primarily captures global variance but does not explicitly model local neighborhoods, and over t-SNE (Van der Maaten & Hinton, 2008), which strongly preserves local structure but may distort broader global relationships.

In the context of narrative analysis, this dimensionality reduction step ensures that semantically similar documents remain geometrically close after projection, facilitating the formation of coherent topic clusters in subsequent stages.

As in the previous section, all notation introduced here is explicitly defined. In particular, $\mathbf{x}_i \in \mathbb{R}^H$ denotes the high-dimensional embedding, $\mathbf{z}_i \in \mathbb{R}^d$ the reduced representation, and v_{ij}, w_{ij} the pairwise membership strengths.

2.3.3 Stage 3: Density-Based Clustering via HDBSCAN

With documents projected into a low-dimensional semantic space, BERTopic applies Hierarchical Density-Based Spatial Clustering of Applications with Noise (HDBSCAN) (Campello et al., 2013) to partition them into coherent topical groups. HDBSCAN identifies clusters of arbitrary shape and varying density without requiring the number of clusters to be specified in advance, a fundamental requirement when the thematic structure of a corpus is unknown *a priori*.

Let $\{\mathbf{z}_i \in \mathbb{R}^d\}_{i=1}^N$ denote the set of low-dimensional document embeddings obtained from the previous stage. HDBSCAN begins by computing the *core distance* of each point $\mathbf{z}_i$ with respect to a minimum neighborhood size k:

$$\mathrm{core}_k(\mathbf{z}_i) = d\left(\mathbf{z}_i, \mathbf{z}_i^{(k)}\right), \tag{2.9}$$

where $\mathbf{z}_i^{(k)}$ denotes the k-th nearest neighbor of $\mathbf{z}_i$ in $\mathbb{R}^d$, and $d(\cdot,\cdot)$ denotes the distance metric used in the reduced space, typically Euclidean distance.

Pairwise distances are then transformed into *mutual reachability distances*, which make the clustering process more robust to variations in local density:

$$d_{\text{mreach}}(\mathbf{z}_i, \mathbf{z}_j) = \max\left\{\mathrm{core}_k(\mathbf{z}_i), \mathrm{core}_k(\mathbf{z}_j), d(\mathbf{z}_i, \mathbf{z}_j)\right\}. \tag{2.10}$$

This transformation stabilizes the distance metric in sparse regions of the embedding space, preventing low-density areas from fragmenting meaningful clusters. HDBSCAN then constructs a minimum spanning tree over the mutual reachability graph, extracts a condensed cluster hierarchy by progressively increasing a density threshold, and selects the partition that maximizes cluster stability.

A distinctive and practically important property of HDBSCAN is its treatment of outliers: points that do not belong to any sufficiently dense region are assigned to a noise class rather than forced into the nearest cluster. In the context of news media corpora, where many documents correspond to isolated or highly specific events, this property improves the coherence and interpretability of the resulting topics by preventing unrelated documents from contaminating well-defined clusters.

As in the previous stages, all notation is explicitly defined. In particular, $\mathbf{z}_i \in \mathbb{R}^d$ denotes the reduced embedding of document i, $\mathrm{core}_k(\cdot)$ denotes the core distance function, and $d_{\text{mreach}}(\cdot,\cdot)$ denotes the mutual reachability distance.

2.3.4 Stage 4: Topic Representation via Class-Based TF-IDF

The final stage translates the geometric clusters produced by HDBSCAN into linguistically interpretable topic representations. BERTopic employs a class-based extension of the classical Term Frequency–Inverse Document Frequency (TF-IDF) scheme. The key conceptual innovation lies in the unit of analysis: rather than computing term weights at the document level, all documents assigned to a given cluster c are concatenated into a single macro-document, and TF-IDF is computed at the cluster level. This reframing shifts the objective from identifying terms important within a document to identifying terms that are characteristic of a topic.

Let $\mathcal{V}$ denote the vocabulary of the corpus and let c index the clusters obtained in the previous stage. The class-based TF-IDF weight of term $t \in \mathcal{V}$ in class c is defined as:

$$W_{t,c} = tf_{t,c} \times \log\left(1 + \frac{A}{tf_t}\right), \tag{2.11}$$

where $tf_{t,c}$ denotes the frequency of term t within cluster c, tf_t denotes the total frequency of term t across all clusters, and A denotes the average number of words per cluster. The logarithmic inverse-frequency term down-weights corpus-wide high-frequency terms analogously to standard IDF, while the additive smoothing constant ensures numerical stability when tf_t is small. From a statistical perspective, this formulation acts as a contrastive measure of term salience: a term receives a high weight in class c if it is disproportionately frequent in that class relative to its overall frequency in the corpus. Consequently, the resulting ranked vocabulary provides a human-interpretable summary of each topic's semantic content, effectively bridging the geometric structure identified by HDBSCAN with the linguistic representation required for analysis.

As in the previous stages, all notation is explicitly defined. In particular, $tf_{t,c}$ and tf_t represent class-specific and global term frequencies, respectively, and $W_{t,c}$ denotes the resulting term weight.

2.3.5 The BERTopic Pipeline as a Statistical Framework

The four stages of BERTopic form a coherent statistical framework for transforming raw text into interpretable narrative topics. SBERT defines a high-dimensional semantic space in which similarity between documents is encoded geometrically; UMAP projects this representation into a lower-dimensional space that facilitates clustering while preserving local structural relationships; HDBSCAN partitions this space into stable, density-based groups without requiring a predefined number of clusters; and c-TF-IDF translates these groups into ranked vocabularies that are interpretable to human analysts.

This pipeline directly addresses several limitations of Latent Dirichlet Allocation discussed in the previous section. It operates in a continuous semantic space rather than a discrete count-based representation; it does not rely on the bag-of-words assumption; it infers the number of topics from the data rather than requiring it as an input parameter; and it explicitly accounts for noise and heterogeneity in large-scale text corpora through density-based clustering.

From a statistical perspective, the pipeline can be interpreted as a sequence of transformations across different representational spaces:

$$d_i \longrightarrow \mathbf{x}_i \in \mathbb{R}^H \longrightarrow \mathbf{z}_i \in \mathbb{R}^d \longrightarrow c_i \longrightarrow W_{t,c}.$$

Here, d_i denotes the original textual document, $\mathbf{x}_i$ its high-dimensional semantic embedding, $\mathbf{z}_i$ the reduced representation obtained after dimensionality reduction, c_i the cluster assignment produced by HDBSCAN, and $W_{t,c}$ the c-TF-IDF weight assigned to term t within cluster c. Each stage preserves specific structural properties while enabling the subsequent transformation, resulting in a consistent mapping from unstructured text to structured thematic representations.

Moreover, the modular architecture of BERTopic allows each component to be independently modified or extended. For example, alternative embedding models or large language models may be incorporated for improved semantic representation or topic labeling, and temporal extensions can be introduced for dynamic corpora.

This flexibility makes BERTopic a robust and extensible framework for computational narrative analysis at scale.

Chapter 3

Methodology

This study adopts a bibliometric and embedding-based topic modeling approach to identify and analyze narrative patterns in news media. The proposed methodology integrates natural language processing techniques with unsupervised machine learning, using the BERTopic framework to extract latent thematic structures from a corpus of documents related to storytelling and narrative in news contexts. The overall workflow consists of data collection, text preprocessing, semantic embedding generation, dimensionality reduction, clustering, and topic representation, followed by qualitative evaluation and interpretation.

3.1 Data Collection and Search Strategy

The dataset was retrieved from the Scopus database using a structured search strategy. The query was designed to capture scientific publications related to storytelling and narrative in news media, incorporating terms associated with narrative structures, framing, and journalistic discourse. The search was applied to the title, abstract, and keyword fields of all records indexed in Scopus, without restriction on document type or publication year, to capture the full historical scope of the literature.

The Scopus query was formulated as follows:

```
TITLE-ABS-KEY ( ( ‘‘storytelling’’ OR ‘‘narrative*’’ ) AND ( ‘‘news
media*’’ OR ‘‘news article*’’ OR ‘‘news homepage*’’ ) ) AND PUBYEAR <
2027
```

The wildcard operator * was applied to the term `narrative` to retrieve its morphological and lexical variants, including *narratives*, *narrative-based*, *narrative framing*, and related expressions. In addition, the same operator was applied to `news media`, `news article`, and `news homepage` to capture both singular and plural forms, as well as compound variations commonly indexed in Scopus.

The search was conducted on May 20th, 2026, yielding an initial set of 1,890 records prior to screening.

3.1.1 PRISMA-Based Screening and Data Cleaning

To ensure transparency and reproducibility in the construction of the analytical corpus, the dataset was processed using a PRISMA-adapted screening protocol (Page et al., 2021). In this study, PRISMA was not employed as a systematic review framework in the strict sense, but rather as a structured reporting procedure to document how the raw Scopus export was transformed into a consistent corpus suitable for bibliometric analysis and topic modeling.

The screening process included four main stages: identification, screening, eligibility assessment, and final inclusion. During the identification stage, all records retrieved from Scopus were exported with their available bibliographic metadata. The screening stage involved the detection of duplicate records and the removal of evident noise. During the eligibility stage, the consistency of metadata fields were evaluated, including title, authorship, publication year, language, abstract availability, document type, DOI, keywords, affiliations, and cited references. Finally, the inclusion stage produced the definitive analytical corpus used in the subsequent bibliometric and topic-modeling analyses.

The cleaning procedure prioritized corpus homogeneity and analytical validity. Only English-language journal articles with usable abstracts were retained, whereas non-article document types, records with insufficient textual information, and entries incompatible with the bibliometric workflow were excluded. Records published in 2026 were retained to preserve the complete temporal coverage of the Scopus search. Documents without DOI were not automatically removed; instead, they were flagged during quality control, since DOI absence does not necessarily invalidate a bibliographic record. Similarly, records without cited references were retained for descriptive analyses but excluded from procedures requiring reference data, such as co-citation and bibliographic coupling.

The detailed PRISMA-adapted flow, including the number of records retained and excluded at each stage, is presented in Figure 4.1 and discussed in the results chapter as part of the corpus construction and document selection analysis.

3.2 Analytical Tools

The bibliometric analyses were conducted using the `bibliometrix` package for R (Aria & Cuccurullo, 2017) and VOSviewer. The `bibliometrix` package was used to compute descriptive indicators and perform bibliometric analyses, including co-citation, bibliographic coupling, and thematic analyses, whereas VOSviewer was employed for the visualization of selected bibliometric networks and exploratory science mapping analyses. All analyses were conducted in R version 4.4.1 to ensure reproducibility.

3.3 Bibliometric Analysis Framework

The study employed several standard bibliometric techniques to examine the social, intellectual, and conceptual structure of the literature. Co-authorship analysis was used to identify collaborative relationships among authors, institutions, and countries. Co-citation analysis examined the intellectual structure of the field by identifying references that are frequently cited together, whereas bibliographic coupling was applied to detect related documents sharing common references and to identify emerging research fronts. In addition, keyword co-occurrence analysis was conducted to explore the conceptual structure of the literature and identify the main thematic areas and their interrelationships within the corpus.

3.4 Topic Modeling Implementation

Topic modeling was implemented using BERTopic (Grootendorst, 2022) to identify latent thematic structures within the analytical corpus.

Figure 3.1 summarizes the workflow used in this stage. Minimal preprocessing was applied before embedding generation to preserve the natural linguistic structure required by transformer-based models, whereas lexical cleaning was reserved for topic representation.

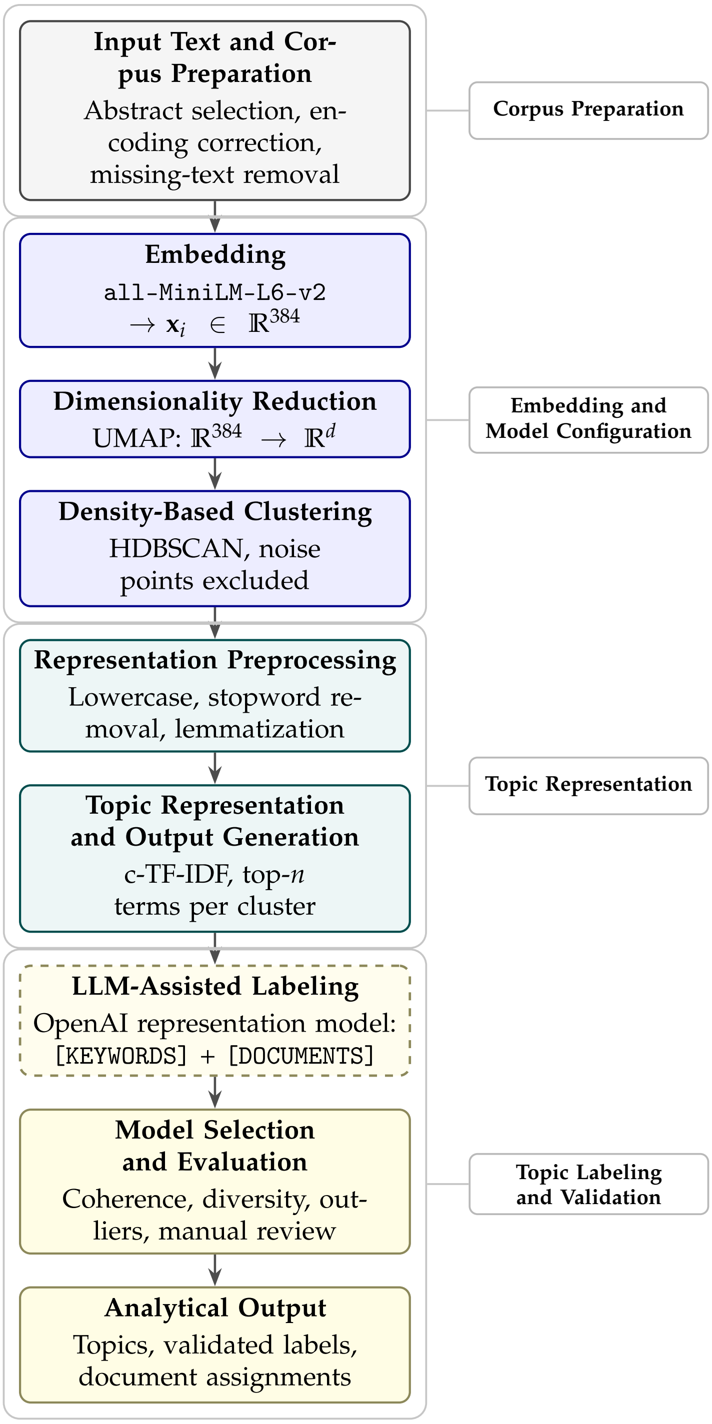


FIGURE 3.1: Workflow of the BERTopic framework used for topic modeling. Colors indicate the functional stages of the workflow: corpus preparation, topic creation, topic representation, and topic labeling and validation. The dashed box indicates the post-clustering OpenAI-assisted labeling stage. Source: Authors' own elaboration.[1]

3.4.1 Input Text and Corpus Preparation

The analysis used the final corpus obtained after the bibliometric screening and cleaning process. For the topic modeling stage, only the abstracts available in the Scopus records were used as textual input. Titles and keywords were retained as bibliographic metadata to support interpretation and validation, but they were not concatenated with the abstracts for model estimation. Preprocessing before embedding generation was limited to correcting encoding inconsistencies, removing records without usable abstracts, normalizing whitespace, and standardizing minor formatting issues.

3.4.2 Embedding and Model Configuration

Embedding

Document embeddings were generated using the `all-MiniLM-L6-v2` model from the SentenceTransformers framework.[2] This model was selected after comparing alternative SentenceTransformer embeddings within the BERTopic pipeline. In particular, `all-MiniLM-L6-v2` was evaluated against `all-mpnet-base-v2` using topic coherence, topic diversity, outlier ratio, and qualitative inspection of representative abstracts.

The final configuration was retained because it provided the most suitable balance between interpretability, computational feasibility, and reproducibility for the abstract-level corpus of 1,294 documents. Although `all-mpnet-base-v2` offers higher-dimensional embeddings, the selected `all-MiniLM-L6-v2` configuration produced a more suitable topic structure for the purposes of this study. More specialized embedding models, such as `BGE` models or scientific-document embedders such as SPECTER, are left for future extensions.

Dimensionality Reduction

Dimensionality reduction was performed using UMAP. This step projects the 384-dimensional document embeddings into a lower-dimensional space that is more suitable for density-based clustering.

The hyperparameters evaluated include `n_neighbors`, which indicates the balance between preserving local and broader neighborhood structures. Smaller values emphasize local semantic proximity, whereas larger values capture a wider topological representation of the corpus. The `n_components` parameter defines the dimensionality of the reduced space passed to HDBSCAN. Furthermore, `min_dist` controls how tightly points can be arranged in the reduced space, and the distance `metric` determines how similarity between embeddings is computed. These parameters were included in the model selection process because changes in the reduced geometry can affect the number, size, and separation of the resulting topics.

[1] d denotes the number of dimensions retained after UMAP reduction ($d = 5$, configurable), and n represents the number of top representative terms extracted per topic using c-TF-IDF. The OpenAI representation model was used only after topic generation to support topic labeling from representative terms and documents.

[2] For further details on the embedding model, see the `all-MiniLM-L6-v2` *Hugging Face model card* and the *SentenceTransformers pretrained models documentation*.

Clustering

Document clusters were identified using HDBSCAN. Unlike clustering methods that require the number of groups to be specified in advance, HDBSCAN identifies topics from dense regions of the reduced embedding space.

The most relevant HDBSCAN hyperparameters are `min_cluster_size` and `min_samples`. The first defines the minimum number of documents required for a group to be considered a topic. Larger values tend to produce fewer and broader topics, while smaller values may produce more granular clusters. The second controls the strictness of cluster formation and influences the number of documents assigned to the outlier class.

3.4.3 Representation Preprocessing

Once the document clusters had been identified, an additional preprocessing step was applied to improve topic interpretability in the c-TF-IDF representation. This procedure included lowercasing, punctuation and number removal, stopword removal, and lemmatization using `spaCy` (`en_core_web_sm`). Table 3.1 summarizes the preprocessing stages applied during topic representation.

TABLE 3.1: Summary of representation preprocessing stages and their effect on a fragment.

Stage	Output fragment
Original text	*"This Study examines Agenda-Setting effects in Online News platforms (2019–2023), proposing a data-driven framework for narrative analysis."*
1. Lowercasing	`this study examines agenda-setting effects in online news platforms (2019-2023), proposing a data-driven framework for narrative analysis.`
2. Punctuation and numerical removal	`this study examines agenda-setting effects in online news platforms proposing a data-driven framework for narrative analysis`
3. Stopword removal	`agenda-setting effects online news platforms data-driven framework narrative analysis`
4. Lemmatization	`agenda-setting effect online news platform data-driven framework narrative analysis`

3.4.4 Topic Representation and Output Generation

After the preprocessed text, c-TF-IDF was used to extract the most representative terms for each topic. The outputs included topic labels, representative keywords, topic frequencies, document-topic assignments, and visualizations of the topic structure.

3.4.5 LLM-Assisted Labeling

With the topic structure already defined, the final stage of the workflow focused on topic representation, labeling, and interpretive validation. At this point, the topics had already been formed through the embedding, UMAP, and HDBSCAN stages.

Therefore, the labeling procedure did not change the number of topics, the cluster structure, or the document-topic assignments.

To support the formulation of concise and academically readable topic labels, an LLM-assisted representation step was incorporated using the OpenAI representation model available in BERTopic.[3] Specifically, the labeling procedure was implemented using `gpt-4o-mini`[4], following BERTopic's LLM-based topic representation workflow (Grootendorst, 2022). For each topic, the prompt included the topic identifier, the representative c-TF-IDF terms `[terms]`, and a small set of representative abstracts `[documents]`. The model was instructed to propose a short academic label using only the supplied evidence and without introducing unsupported concepts. The methodological prompt used for this stage was defined as follows:

TABLE 3.2: Prompt template used for OpenAI-assisted topic labeling.

Prompt template

```
You are assisting with academic topic labeling for a research
project about storytelling, narratives, and news media.
Use only the evidence provided below.  Do not introduce concepts
that are not supported by the representative terms or abstracts.
Topic ID: {topic_id}
Representative c-TF-IDF terms:  {terms}
Representative abstracts:  {documents}
Return the answer exactly in this format:
topic:  <short academic label>
rationale:  <one concise sentence explaining the label>
```

The labels generated by `gpt-4o-mini` were treated as preliminary suggestions rather than final outputs. Each proposed label was manually reviewed by comparing it with the representative terms, the selected abstracts, and the broader research objective of the study. Labels were adjusted when they were too broad, too narrow, or not sufficiently aligned with the dominant content of the topic. Thus, the final topic names reported in the results correspond to researcher-validated labels supported by c-TF-IDF evidence, representative abstracts, and LLM-assisted linguistic refinement.

TABLE 3.3: Planned structure for LLM-assisted topic labeling.[5]

Element	**Role in the labeling procedure**
c-TF-IDF terms	Provide the initial statistical representation of each topic by identifying the most representative terms within each cluster. These terms are passed to the prompt through the `[KEYWORDS]` field.
Representative abstracts	Provide contextual evidence for interpreting the meaning of the topic beyond isolated keywords. These abstracts are passed to the prompt through the `[DOCUMENTS]` field.

[3]For further details, see the BERTopic LLM representation docs.

[4]For information on the model used in this study, see the OpenAI GPT-4o mini model docs.

Element	**Role in the labeling procedure**
OpenAI representation model	Used as a linguistic support tool to propose concise academic labels from the representative terms and abstracts. The model is incorporated through BERTopic's `OpenAI` representation model.
Prompt structure	Defines the evidence shown to the model and constrains the output so that the label is based only on the supplied information. The expected output follows the format `topic: <topic label>`.
Researcher validation	Ensures that the proposed label is consistent with the representative terms, the selected abstracts, and the research objective before being accepted, revised, or replaced.

TABLE 3.4: Illustrative example of OpenAI-based topic representation in BERTopic.[6]

Default representation		**OpenAI representation**
climate | *environmental* | *energy*	$\longrightarrow$	Environmental Narratives and Media Framing
trump | *election* | *scandal*	$\longrightarrow$	Trust and Objectivity in News Media
china | *chinese* | *geopolitical*	$\longrightarrow$	Chinese Media and Geopolitical Narratives

3.4.6 Model Selection and Evaluation

Model selection was based on both quantitative and qualitative criteria, including topic coherence (c_v), topic diversity, outlier ratio, silhouette score, interpretability of topic labels, and thematic consistency of representative documents.

First, topic coherence was computed using the c_v measure. This metric evaluates whether the most representative terms of a topic are semantically related and tend to occur in meaningful contexts within the corpus. Let $W_k^{(n)} = \{w_{k1}, \ldots, w_{kn}\}$ denote the set of the top-n representative terms for topic k. In general terms, topic coherence can be written as

$$C_v(k) = \text{Coherence}\left(W_k^{(n)}, \mathcal{D}\right), \tag{3.1}$$

where $\mathcal{D}$ denotes the document collection used to estimate word co-occurrence patterns. The c_v measure is commonly used because it combines indirect confirmation measures with normalized co-occurrence information (Röder, Both, & Hinneburg, 2015).

[5]Table 3.3 describes the planned structure for the LLM-assisted labeling stage. The procedure follows the logic described in the official BERTopic LLM representation docs, where OpenAI can be used as a representation model to transform topic keywords and representative documents into readable topic labels.

[6]The examples illustrate how representative c-TF-IDF terms can be converted into concise academic topic labels through an OpenAI-assisted representation step. The labels do not modify the BERTopic clusters; they only support interpretation after the topics have been estimated.

Second, topic diversity was used to assess the degree of lexical overlap across topics. This metric captures whether the model produces distinct thematic structures or whether several topics rely on nearly the same representative terms. Following the usual definition, topic diversity is computed as

$$TD = \frac{\left|\bigcup_{k=1}^{K} W_k^{(n)}\right|}{K \times n}, \tag{3.2}$$

where K is the number of substantive topics and n is the number of top terms considered per topic. Values closer to 1 indicate that the topics share fewer top terms and are therefore less redundant (Dieng, Ruiz, & Blei, 2020).

Third, the outlier ratio was examined because BERTopic relies on HDBSCAN, a density-based clustering algorithm that does not force every document into a cluster. Documents that do not belong to a sufficiently dense region are assigned to the outlier class, usually denoted as -1 (Campello et al., 2013). The outlier ratio was computed as

$$OR = \frac{N_{-1}}{N}, \tag{3.3}$$

where N_{-1} is the number of documents assigned to the outlier class and N is the total number of modeled documents. A very high outlier ratio may suggest that the clustering configuration is too restrictive, whereas a very low value may indicate that heterogeneous documents are being forced into topics.

Finally, the silhouette score was considered as an additional diagnostic of cluster separation. For a document i, the silhouette value compares its average distance to documents in the same cluster with its average distance to documents in the nearest alternative cluster:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}}, \tag{3.4}$$

where $a(i)$ is the average distance between document i and the other documents assigned to the same cluster, and $b(i)$ is the average distance between document i and the documents in the nearest alternative cluster. Values closer to 1 indicate better separation between clusters, while values near 0 suggest overlapping cluster boundaries (Rousseeuw, 1987).

The representative c-TF-IDF terms, high-probability abstracts, and outlier documents were reviewed to determine whether the topics were interpretable in relation to the research objective. Therefore, the final model was selected by balancing numerical diagnostics with thematic coherence and interpretive validity.

3.4.7 Analytical Use of the Topics

The extracted topics were interpreted as latent thematic patterns within the literature on storytelling, narratives, and news media. The analysis considered topic prevalence, semantic relationships among topics, and their contribution to the study objectives.

Chapter 4

Results

This chapter presents the results obtained from the final corpus of 1,294 journal articles retrieved from Scopus. The analysis begins with the corpus construction process, using a PRISMA-adapted workflow to document the identification, screening, eligibility, and inclusion stages that led to the final analytical dataset. Then, it moves to the bibliometric analysis, outlining the general profile of the corpus and examining patterns of scientific production, source distribution, country participation, and keyword co-occurrence. These results provide the broader scholarly context in which research on storytelling, narratives, and news media has evolved.

The final part of the chapter focuses on the abstract-based BERTopic analysis. It reports the model configuration, embedding and hyperparameter selection, diagnostic metrics, outlier review, topic-labeling procedure, and the interpretation of the narrative themes identified in the corpus.

4.1 Corpus Construction

The initial Scopus search retrieved 1,890 records. A cleaning protocol was subsequently applied to ensure bibliographic consistency, textual completeness, and document-type homogeneity. Figure 4.1 presents the PRISMA-adapted workflow used to construct the final corpus, while Table 4.1 summarizes the step-by-step reduction process.

TABLE 4.1: Step-by-step corpus reduction from the initial Scopus export to the final analytical corpus.

Step	Criterion	Excluded	Remaining	Rationale
0	Initial Scopus export	–	1,890	Records retrieved using the final Scopus search query.
1	Duplicate and noise control	30	1,860	Removal of duplicate records, non-eligible document noise, and one book-review-like record.
2	Language and abstract screening	49	1,811	Retention of English records with usable abstracts.
3	Document-type harmonization	515	1,296	Retention of journal articles only to ensure bibliometric and textual homogeneity.

Step	Criterion	Excluded	Remaining	Rationale
4	Final metadata reconciliation	2	1,294	Alignment with the canonical Scopus file enriched with cited references.
Total	**Records excluded**	**596**	**1,294**	Final corpus used for the bibliometric analysis.

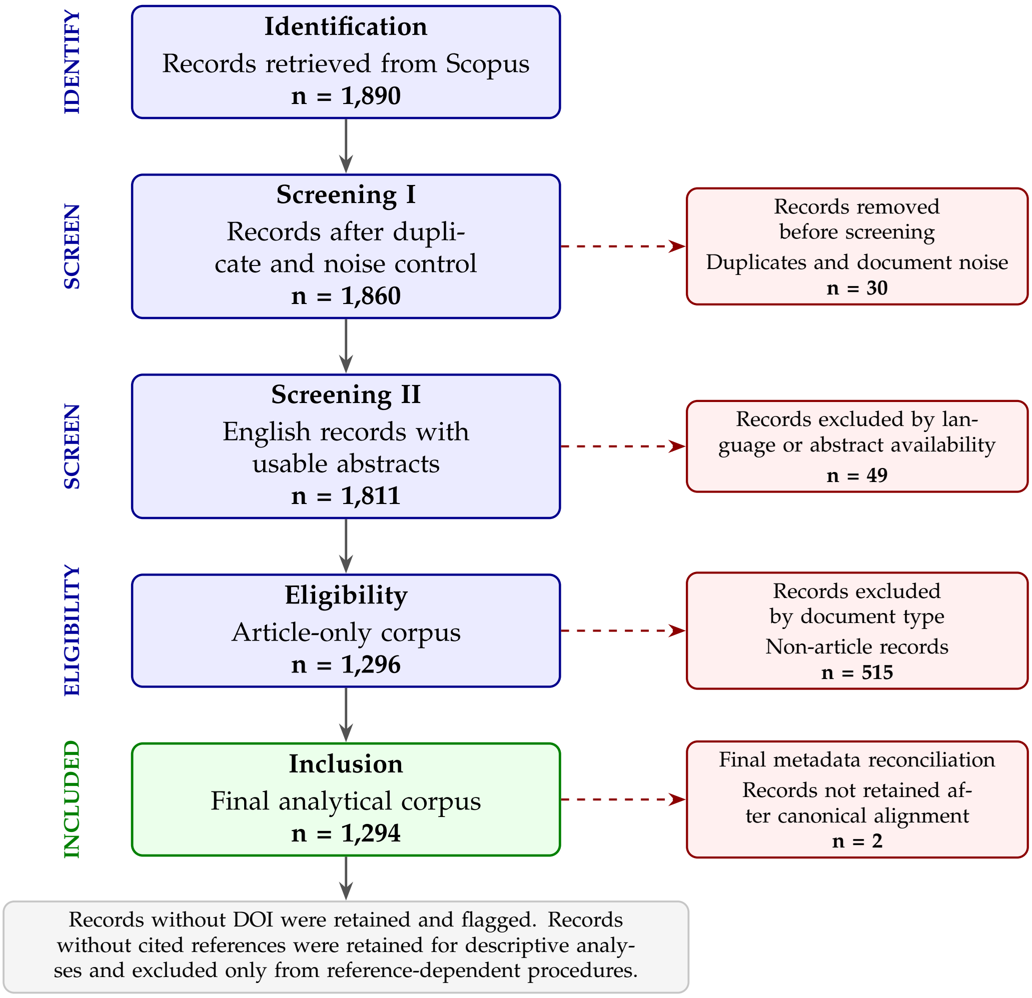


FIGURE 4.1: PRISMA-adapted workflow for corpus construction and document selection. Adapted from Page et al. (2021). Source: Authors' own elaboration.

The final corpus comprises 1,294 English-language journal articles published between 1989 and 2026. Most records include DOI information (1,251 documents) and cited references (1,267 documents), providing sufficient coverage for both performance analysis and science mapping.

4.2 Bibliometric Analysis

4.2.1 General Information

Table 4.2 summarizes the main bibliometric characteristics of the corpus, including productivity, citation impact, document contents, and collaboration patterns. The results indicate sustained growth in research on storytelling, narratives, and news media, together with increasing thematic diversification and moderate levels of international collaboration. All documents in the corpus correspond to journal articles, ensuring document-type homogeneity for the subsequent analyses.

TABLE 4.2: Main information about the bibliographic dataset.[1]

Description	Results
Main information about data	
Timespan	1989–2026
Sources (journals, books, etc.)	725
Documents	1,294
Annual growth rate (%)	14.52
Document average age	5.02
Average citations per document	12.86
References	70,549
Document contents	
Keywords Plus (ID)	2,218
Author's Keywords (DE)	3,986
Authors	
Authors	2,985
Authors of single-authored documents	427
Authors collaboration	
Single-authored documents	436
Co-authors per document	2.48
International co-authorships (%)	18.62
Document types	
Article	1,294

Overall, the corpus reflects an active and expanding research field combining journalism, media studies, digital communication, and computational text analysis.

Figure 4.2 shows the temporal evolution of scientific production and mean citations per year. Publication activity remained relatively limited until the early 2000s, followed by sustained growth after 2010 and a marked acceleration from 2020 onward. This trend suggests increasing scholarly interest in storytelling, narratives, and news media research. The apparent decline observed in 2026 should be interpreted cautiously, since data collection was conducted before the end of the publication year.

[1]Annual growth rate denotes the average yearly increase in scientific production over the study period. Document average age refers to the average number of years since publication. Keywords Plus are index terms generated by the database from cited references, whereas Author's Keywords are provided by the authors. Co-authors per document and international co-authorships describe the collaborative structure of the corpus.

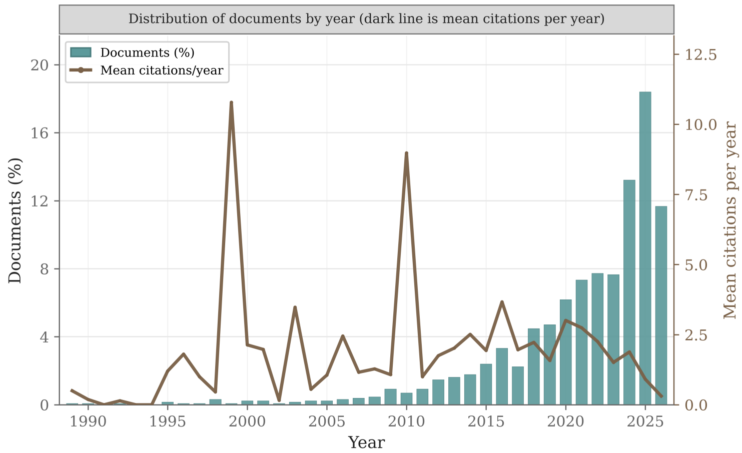
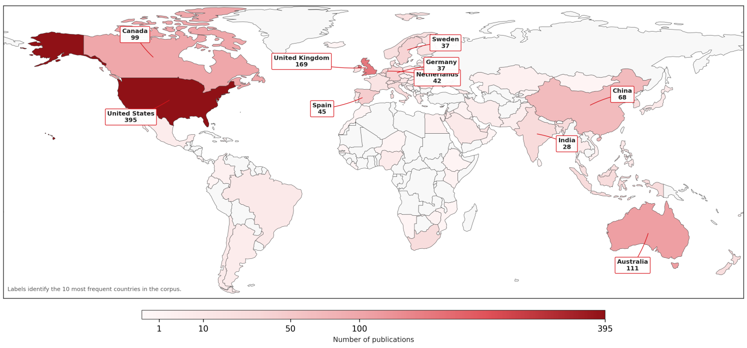


FIGURE 4.2: Annual scientific production and mean citations per year. Source: Authors' own elaboration based on Scopus data.

4.2.2 Geographical Distribution and Country Productivity

The geographical distribution of scientific production shows a strong concentration in English-speaking research systems. As shown in Figure 4.3, the United States is the leading contributor, with 395 publications associated with the corpus, followed by the United Kingdom, Australia, Canada, and China.



R	Country	Pub.
1	United States	395
2	United Kingdom	169
3	Australia	111
4	Canada	99
5	China	68
6	Spain	45
7	Netherlands	42
8	Sweden	37
9	Germany	37
10	India	28

FIGURE 4.3: Geographical distribution of scientific production and top 10 most productive countries.

This distribution also suggests that the internationalization of the field is uneven. Whereas the corpus includes contributions from different regions, the highest levels of production remain concentrated in a relatively small group of countries. This pattern is relevant because research on news media narratives is closely linked to language, media systems, and national journalistic traditions; therefore, the geographical concentration of publications may also influence the types of narratives, cases, and media contexts most frequently studied.

4.2.3 Keyword Co-occurrence Network

Figure 4.4 shows a dense conceptual network organized around *news media*, which appears as the largest and most central node. This position indicates that the term functions as the main conceptual anchor of the corpus, connecting different lines of research related to journalism, discourse, digital platforms, and narrative analysis.

The thickness of the links suggests that several of these concepts tend to appear together rather than as isolated thematic areas.

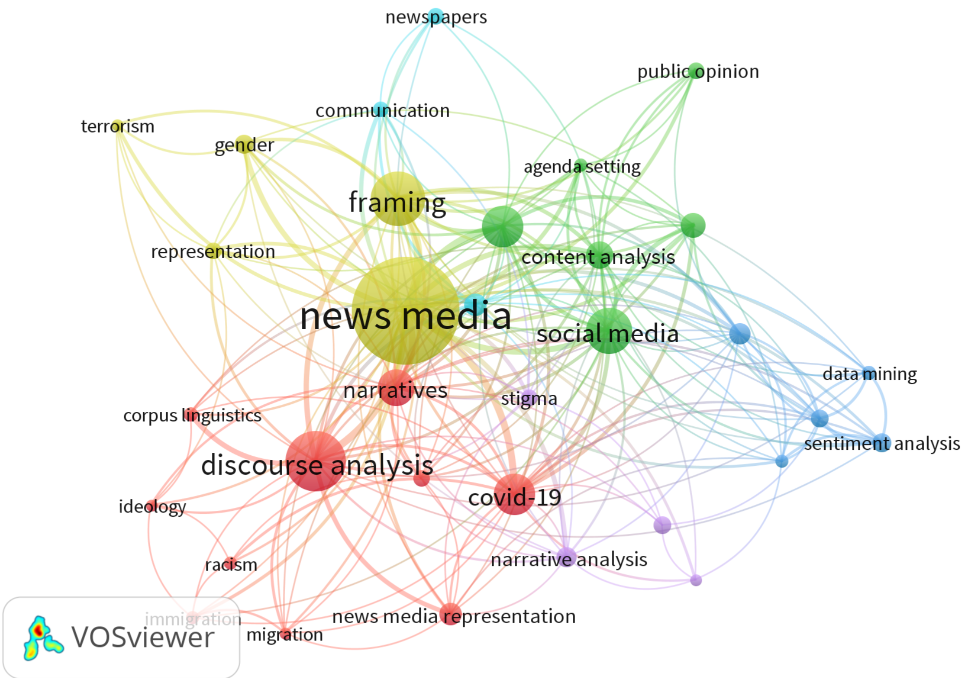


FIGURE 4.4: Keyword co-occurrence network.

The yellow cluster is mainly articulated around *news media*, *framing*, *gender*, *representation*, and *terrorism*. This group points to studies concerned with how news organizations frame social and political issues, and how media narratives shape public representations of conflict, identity, and social problems.

The red cluster is centered on *discourse analysis*, *narratives*, *covid-19*, *migration*, *racism*, and *news media representation*. This community reflects a more interpretive and critical orientation, where keywords are associated with the analysis of language, representation, crisis narratives, and marginalized groups in news coverage.

The green cluster brings together terms such as *social media*, *content analysis*, *agenda setting*, and *public opinion*. These associations suggest a line of work focused on the interaction between news circulation, digital platforms, and public perception. In this area, narrative and media effects are often examined through the relationship between online communication and audience interpretation.

The blue cluster includes *sentiment analysis*, *data mining*, and related computational terms. Although smaller than the central media and discourse clusters, it shows the presence of methodological approaches that use automated text analysis to study patterns in media content. Finally, the purple cluster, located closer to *narrative analysis* and *stigma*, suggests a more specialized area concerned with the interpretation of narratives in sensitive social or health-related contexts.

Overall, the network suggests that the literature is structured around two closely connected poles: one focused on news media, framing, and public communication, and another centered on discourse, representation, and narrative interpretation. Around these poles, computational methods and platform-based communication appear as complementary areas that extend the study of news media narratives toward large-scale textual analysis.

4.2.4 Model Configuration and Selection

Before estimating the final topic structure, several BERTopic configurations were evaluated by combining two SentenceTransformer embedding models with different UMAP and HDBSCAN parameter settings. The comparison was based on the evaluation criteria defined in the methodology: topic coherence, topic diversity, outlier ratio, and silhouette score. These metrics were considered jointly, rather than as isolated indicators, because the objective was to select a model that produced interpretable topics whereas maintaining a reasonable level of thematic separation and avoiding an excessive number of outlier documents.

TABLE 4.3: BERTopic model selection based on embedding and clustering configurations.[2]

Run	Embedding model	Topics	Outliers (%)	Topic diversity	c_v	Silhouette
all-MiniLM-L6-v2_g1	`all-MiniLM-L6-v2`	**25**	**15.53**	**0.9640**	**0.4506**	**0.5220**
all-MiniLM-L6-v2_g2	`all-MiniLM-L6-v2`	29	24.73	0.9552	0.4478	0.5756
all-mpnet-base-v2_g1	`all-mpnet-base-v2`	22	16.15	0.9500	0.4472	0.5291
all-MiniLM-L6-v2_g4	`all-MiniLM-L6-v2`	16	34.16	0.9750	0.4073	0.5377
all-MiniLM-L6-v2_g3	`all-MiniLM-L6-v2`	23	25.66	0.9522	0.4317	0.5139
all-mpnet-base-v2_g4	`all-mpnet-base-v2`	15	30.29	0.9467	0.4180	0.4873
all-mpnet-base-v2_g3	`all-mpnet-base-v2`	18	21.33	0.9611	0.4027	0.4759
all-mpnet-base-v2_g2	`all-mpnet-base-v2`	23	18.16	0.9348	0.3983	0.5421

Table 4.3 summarizes the configurations tested. The selected configuration was `all-MiniLM-L6-v2_g1`, which used `all-MiniLM-L6-v2` as the embedding model. Table 4.4 details the final BERTopic configuration selected for the abstract-level topic modeling analysis.

TABLE 4.4: Selected BERTopic model configuration.[3]

Parameter / Metric	**Value**
Selected model	
Run ID	`all-MiniLM-L6-v2_g1`
Embedding model	`all-MiniLM-L6-v2`
Embedding dimension	384
Documents modeled	1,294
UMAP dimensionality reduction	
Number of neighbors	10
Number of components	5
Minimum distance	0.00
Distance metric	Cosine
HDBSCAN clustering	
Minimum cluster size	15
Minimum samples	5
Distance metric	Euclidean
Topic structure	
Topics identified	25

[2]The highlighted row indicates the selected configuration. Model selection was based on the criteria defined in the methodology: topic coherence, topic diversity, outlier ratio, silhouette score, and qualitative inspection of representative abstracts.

Parameter / Metric	**Value**
Outlier documents	201
Model evaluation metrics	
Outlier ratio (%)	15.53
Topic diversity (TD)	0.964
Topic coherence c_v	0.4506
Silhouette score (S)	0.522

This configuration obtained the highest topic coherence value and the lowest outlier ratio among the best-performing models, while still producing a sufficiently detailed topic structure.

Although `all-MiniLM-L6-v2_g2` obtained the highest silhouette score, it also produced a higher number of topics and a considerably larger proportion of outlier documents. In contrast, `all-MiniLM-L6-v2_g1` offered a more balanced solution: it achieved the highest topic coherence, retained a lower outlier ratio, and produced 25 topics, which allowed a detailed but still interpretable representation of the corpus. For this reason, this configuration was retained as the final model for the BERTopic analysis.

Model Diagnostics

The topic coherence obtained ($c_v = 0.4506$) indicates that the representative terms within each topic maintain a reasonable conceptual relationship, allowing for consistent thematic interpretation. Although the value is moderate, this is expected in an interdisciplinary corpus composed of scientific abstracts from areas such as journalism, media studies, health, politics, environment, and computational methods.

The topic diversity value ($TD = 0.964$) shows that most representative terms are specific to individual topics, suggesting a low level of vocabulary redundancy across clusters. This result supports the differentiation of the thematic structures identified by the model.

The silhouette score ($S = 0.522$) reflects a moderate separation between document clusters in the reduced embedding space. In practical terms, this indicates that the topics show an acceptable degree of internal cohesion and distinction from other groups.

Finally, the outlier ratio was 15.53%, meaning that 201 abstracts were not assigned to any topic. This proportion suggests that the model preserved a moderate set of semantically ambiguous or weakly connected documents outside the main topic structure, rather than forcing them into clusters with limited thematic fit.

4.2.5 Outlier Review

The outlier class was reviewed separately to assess whether the documents excluded from the main topic structure reflected noise, highly specialized cases, or abstracts with weaker semantic proximity to the identified clusters. These documents were retained in the corpus but treated with caution in the interpretation of the thematic structure.

[3]The Run ID is an internal identifier assigned to distinguish the embedding model and hyperparameter configuration evaluated during model selection. In this case, `all-MiniLM-L6-v2_g1` denotes the first hyperparameter configuration evaluated with the `all-MiniLM-L6-v2` embedding model.

TABLE 4.5: Mini-report of BERTopic outlier documents.

Indicator	Result
Outlier documents	201 abstracts, equivalent to 15.53% of the modeled corpus.
Temporal concentration	140 outlier documents were published between 2021 and 2026, representing 69.7% of the outlier set.
Source dispersion	The outliers were distributed across 168 different sources, indicating that they do not originate from a single journal or publication outlet.
Abstract length	The median abstract length was 205 words. Most outliers had standard abstract lengths rather than extremely short or incomplete descriptions.
Metadata completeness	Only 5 records lacked DOI information, while 24 lacked author keywords. However, 148 records had no index keywords, which may reduce their auxiliary bibliographic description.
Observed thematic profile	The outlier set included documents related to health, digital media, governance, identity, crisis communication, environment, culture, and local narratives. This indicates thematic dispersion rather than simple data noise.

The review suggests that the outlier class should not be interpreted as a set of invalid records. Most documents contained usable abstracts and standard bibliographic metadata, but their semantic profiles were not sufficiently close to the dense regions identified by HDBSCAN. In practical terms, these documents appear to represent peripheral, highly specific, or cross-cutting cases within the broader literature on storytelling, narratives, and news media.

Therefore, the outlier documents were retained as part of the bibliographic corpus but excluded from the thematic interpretation of the 25 BERTopic clusters. This decision avoids forcing semantically weak documents into topics where they could reduce interpretability, whereas preserving transparency about the portion of the corpus that falls outside the main thematic structure.

4.2.6 BERTopic Model Output

Table 4.6 presents the 25 topics identified by the selected BERTopic model. The table reports the final OpenAI-assisted topic labels, the number and percentage of documents assigned to each topic, and the representative c-TF-IDF terms used as evidence for topic interpretation.

TABLE 4.6: BERTopic thematic inventory based on abstracts.[4]

Topic (T_i)	Topic Label	Docs. (%)	Representative terms
0	Environmental Narratives and Media Framing	165 (12.75)	climate; environmental; energy; sustainability; nuclear; sustainable; transition; animal
1	Trust and Objectivity in News Media	142 (10.97)	trump; election; scandal; trust; video; virtual; consumption; objectivity
2	Chinese Media and Geopolitical Narratives	99 (7.65)	china; chinese; geopolitical; indonesia; ethnic; boundary; indonesian; chinese news
3	Migration Narratives and Media Representation	78 (6.03)	refugee; immigrant; migration; migrant; border; threat; european; europe
4	Police Violence and Racial Narratives	55 (4.25)	police; crime; mass; murder; black; killing; perpetrate; criminal
5	Public Health Narratives and Media Coverage	51 (3.94)	pandemic; disease; vaccine; outbreak; public health; health; lockdown; health crisis
6	Gender-Based Violence in Media Narratives	43 (3.32)	sexual; violence; girl; victim; feminist; assault; man; abuse
7	Media Representations of Stigma in Substance Use and Mental Health	39 (3.01)	drug; suicide; substance; stigma; medicine; alcohol; intention; screen
8	Disinformation and Strategic Narratives in the Russo-Ukrainian War	35 (2.70)	russian; russia; ukraine; invasion; war; disinformation; strategic narrative; strategic
9	Media Representations of Africa and Conflict	35 (2.70)	africa; african; accountability; church; house; conflict; protection; magazine
10	Financial Sentiment Analysis in News Media	31 (2.40)	sentiment; financial; market; firm; return; real; regression; housing
11	Media Representation of Religious Groups and Terrorism	30 (2.32)	muslim; islamic; terrorist; terrorism; religious; attack; censorship; religion
12	Narratives of Care and Disability in Media	28 (2.16)	disability; care; miss; intellectual; birth; neglect; woman; white woman
13	AI Narratives in Media	28 (2.16)	artificial; artificial intelligence; intelligence; technology; german; automate; fear; autonomous
14	Media Narratives in the Israeli-Palestinian Conflict	26 (2.01)	arabic; israeli; medium politic; conflict; compound; music; relationship medium; middle
15	Gender and Agency in Sports Narratives	25 (1.93)	sport; elite; mother; game; woman; girl; extend; socio cultural

Topic (T_i)	Topic Label	Docs. (%)	Representative terms
16	Information Extraction and Analysis in News Media	24 (1.85)	extraction; task; temporal; computational; judge; domain; accuracy; technique
17	Education Policy and Media Representation	22 (1.70)	education; school; reform; university; teaching; educational; conservative; funding
18	Gender and Media Narratives	21 (1.62)	woman; patriarchal; gender; feminist; intersectional; gendere; south africa; empowerment
19	Indian Media and Socio-Political Narratives	21 (1.62)	indian; india; mobile; minor; sea; science; distance; recur
20	Indigenous Narratives and Media Representation	20 (1.55)	indigenous; new zealand; zealand; native; sound; indigenous people; responsive; culturally
21	Weight Loss and Science Communication	20 (1.55)	weight; physical; science; loss; scientific; uncertain; comment; bias
22	Data Visualization in Storytelling	19 (1.47)	visualization; map; chart; interactive; belief; experiment; analytic; recall
23	Media Representation of Marginalized Communities	19 (1.47)	issue affect; spanish language; magazine; marginalized; japan; people news; star; family
24	Urban and Rural Narratives in Media	17 (1.31)	rural; urban; resident; town; developer; industrial; compete narrative; press coverage

The results show that research on storytelling, narratives, and news media is mainly concentrated around environmental communication, trust in news, and geopolitical media narratives. The largest topics correspond to environmental narratives and media framing (T_0, 165 documents), trust and objectivity in news media (T_1, 142 documents), and Chinese media and geopolitical narratives (T_2, 99 documents). These topics suggest that a substantial part of the corpus examines how news media construct public meaning around risk, institutional credibility, political conflict, and international affairs.

In addition, several topics address the representation of social groups and contested public issues. These include migration narratives and media representation (T_3), police violence and racial narratives (T_4), gender-based violence in media narratives (T_6), media representations of stigma in substance use and mental health (T_7), religious groups and terrorism (T_{11}), Indigenous narratives (T_{20}), and marginalized communities (T_{23}). This thematic group indicates that narrative analysis is frequently used to examine visibility, stigma, exclusion, and power relations in media discourse.

[4]Topic labels were generated through the OpenAI-assisted labeling stage described in the methodology. The labeling procedure used representative c-TF-IDF terms and selected abstracts as evidence. These labels support interpretation but do not modify the clusters or document-topic assignments produced by BERTopic. T_i indicates the topic number in the table.

Complementary lines of research also emerge around public health, war, technology, and applied computational methods. Public health narratives and media coverage (T_5), disinformation and strategic narratives in the Russo-Ukrainian War (T_8), AI narratives in media (T_{13}), information extraction and analysis in news media (T_{16}), financial sentiment analysis (T_{10}), and data visualization in storytelling (T_{22}) show that the corpus combines interpretive concerns with computational and data-driven approaches.

Overall, the identified topics indicate that the field is organized around the intersection of media framing, public trust, geopolitical conflict, social representation, and computational analysis. Rather than treating storytelling as a purely stylistic feature of journalism, the corpus positions narrative as a mechanism through which news media structure public understanding of complex social, political, environmental, and technological issues.

4.2.7 Spatial Distribution of Topics

Figure 4.5 shows the spatial distribution of the abstract-level topics obtained with BERTopic. Each point represents one abstract projected into a two-dimensional space using UMAP. Documents located close to one another are interpreted as semantically similar in the reduced embedding space, whereas more distant regions indicate weaker thematic proximity. Colored points correspond to documents assigned to one of the 25 topics, while gray points represent outlier documents not assigned to a dense topic cluster by HDBSCAN.

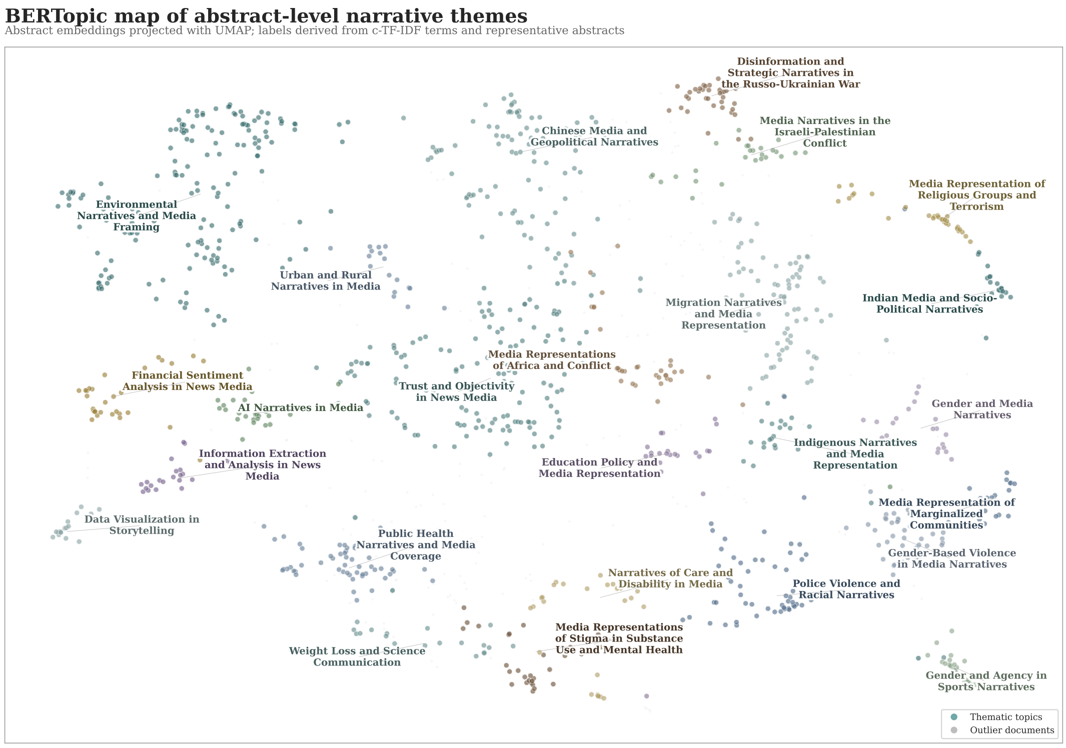


FIGURE 4.5: BERTopic map of abstract-level narrative themes.

The map reveals several differentiated thematic regions. On the left side of the space, environmental narratives, trust and objectivity in news media, financial sentiment analysis, artificial intelligence, information extraction, and data visualization appear as relatively distinct areas. This suggests that the corpus contains both interpretive and computational lines of research, with some topics focused on media framing and others on methodological applications to news content.

The upper and central-right regions are mainly associated with geopolitical and conflict-related narratives, including Chinese media and geopolitical narratives, the Russo-Ukrainian war, the Israeli-Palestinian conflict, migration, and media representations of religious groups and terrorism. These areas indicate that a substantial part of the literature studies how news media construct narratives around international conflict, political identity, borders, and strategic communication.

The lower-right region groups topics related to social representation and inequality, including police violence and racial narratives, gender-based violence, marginalized communities, Indigenous narratives, and gender and agency in sports. The relative proximity of these topics suggests a broader thematic area concerned with visibility, exclusion, stigma, and the media construction of social groups.

Overall, the spatial distribution supports the interpretation that the corpus is organized around three main analytical poles: public risk and media framing, geopolitical conflict and strategic narratives, and social representation and identity. Rather than forming a single compact thematic block, the literature appears as a diversified field in which narrative analysis is applied to environmental, political, social, health-related, and computational domains.

4.2.8 Representative Terms of the Main Topics

Figure 4.6 shows the most representative terms of the main topics according to their c-TF-IDF weights. While Figure 4.5 displayed the semantic proximity among documents and topic regions, this representation highlights the distinctive vocabulary that characterizes each thematic cluster within the corpus.

The results show that several topics are supported by highly specific vocabularies. This is particularly visible in topics such as environmental narratives and media framing (T_0), migration narratives and media representation (T_3), police violence and racial narratives (T_4), public health narratives and media coverage (T_5), and media representation of religious groups and terrorism (T_{11}). In these cases, terms such as *climate*, *refugee*, *police*, *pandemic*, and *muslim* have strong discriminatory value within their respective clusters, indicating thematic areas with clearly identifiable vocabularies.

Other topics, such as trust and objectivity in news media (T_1), financial sentiment analysis in news media (T_{10}), and AI narratives in media (T_{13}), combine more general media-related terms with domain-specific vocabulary. This suggests that part of the corpus is organized around hybrid topics, where broader concerns about journalism and public communication intersect with political, financial, and technological domains.

Taken together, the c-TF-IDF profiles confirm that the BERTopic solution captures both broad narrative structures and more specialized thematic areas. The presence of distinctive terms across the main clusters supports the interpretability of the OpenAI-assisted labels and reinforces the consistency of the topic structure identified in the abstract corpus.

FIGURE 4.6: Representative c-TF-IDF terms for the main BERTopic clusters.

4.2.9 Thematic Evolution and Methodological Convergence

The topics identified in the abstract corpus show that research on storytelling, narratives, and news media has evolved into a diversified field of application. Narrative analysis is not restricted to journalism as a professional practice, but extends to environmental communication, political trust, migration, geopolitical conflict, public health, gender-based violence, financial sentiment, artificial intelligence, and data visualization. This thematic variety shows that narratives operate as analytical devices for examining how news media organize meaning across different areas of public life.

A second important pattern is the increasing presence of computational and embedding-based approaches within the field. Topics related to information extraction, artificial intelligence, financial sentiment analysis, and data visualization suggest that the study of media narratives is progressively being combined with methods from natural language processing and computational social science. In this sense, the results reflect a methodological transition in which interpretive traditions in journalism and media studies coexist with automated techniques for analyzing large-scale textual corpora.

Finally, the topic structure highlights the relevance of public risk and crisis contexts. Environmental narratives, pandemic-related media coverage, migration, war, terrorism, and police violence appear as central domains in which news storytelling is used to frame uncertainty, responsibility, vulnerability, and social conflict. Overall, the BERTopic results suggest a field organized around the convergence of media framing, social representation, crisis communication, and computational text analysis.

Chapter 5

Discussion

The findings of this study should be interpreted in relation to two connected bodies of literature: bibliometric studies that map the structure of scientific fields, and recent applications of embedding-based topic modeling, particularly BERTopic, to large textual corpora. Rather than treating the results as isolated outputs, this discussion examines how the bibliometric and topic modeling findings compare with previous studies, where they converge, where they differ, and what this study adds to the existing literature.

5.0.1 Expansion of the Field and Its Difference from Method-Driven NLP Literature

The bibliometric results show that research on storytelling, narratives, and news media has expanded considerably, reaching 1,294 journal articles in the final corpus. This growth indicates that narrative-oriented media research has become a recognizable scholarly domain, especially in relation to public issues such as environmental communication, political trust, migration, health crises, gendered violence, and geopolitical conflict.

This pattern is related to, but not identical with, the growth described by Warsito, Endro Suseno, and Arifudin (2025). Their systematic review shows that recent work on short-text analysis has been strongly shaped by methodological developments in embeddings, BERT-based representations, and topic modeling. In their case, the field grows around computational problems: how to represent sparse, noisy, and short texts from platforms such as social media. In the present study, by contrast, the growth is not primarily driven by a technical problem, but by a substantive research domain: how news media narratives organize public meaning. This distinction matters because it shows that embedding-based topic modeling is not only relevant for technically oriented NLP corpora, but also for mapping theoretical and applied developments in communication and journalism studies.

The comparison also clarifies why a bibliometric approach is necessary in this thesis. While Warsito et al. (2025) synthesize methodological trends in embedding-based short-text analysis, the present study maps a broader scholarly ecosystem around news media narratives. Therefore, the contribution is not simply to apply BERTopic, but to connect computational topic modeling with the bibliometric structure of a field that remains strongly rooted in media studies, journalism, discourse analysis, and communication research.

5.0.2 A Conceptual Core Anchored in Media Studies, with Computational Expansion

The keyword co-occurrence network shows that the corpus is organized around terms such as *news media*, *framing*, *discourse analysis*, *social media*, *narratives*, *content analysis*, and *covid-19*. This structure suggests that the field has a clear interpretive core: scholars are mainly interested in how news media frame, represent, and circulate meanings around social and political issues.

This finding differs from more technically oriented BERTopic applications. For example, Okazaki and Takahashi (2025) use BERTopic to extract corporate trends from news data and compare those trends with securities reports. In that study, news is treated as a timely source of business information. In this thesis, news media are not treated as raw evidence for external events, but as an object of scholarly interpretation. The focus is not whether news predicts or reflects corporate behavior, but how academic literature studies the narrative role of news in public life.

A similar distinction can be made with the work of Badieah et al. (2024), who use BERTopic for publication venue recommendation. Their objective is practical and predictive: to determine whether topic modeling can support venue classification. In the present study, BERTopic is used for interpretive mapping. The model is not evaluated according to recommendation accuracy, but according to whether it produces coherent, differentiated, and analytically meaningful thematic structures. This difference is important because it explains why metrics such as topic coherence, topic diversity, outlier ratio, and qualitative inspection are more appropriate for this thesis than classification-oriented metrics.

5.0.3 Dominant Topics and the Public Problem Orientation of News Narrative Research

The BERTopic results indicate that the most prominent thematic domains include environmental narratives, trust and objectivity in news media, Chinese geopolitical narratives, migration, police violence, public health, gender-based violence, stigma, war narratives, financial sentiment, religious representation, AI narratives, and data visualization. These topics reveal that the literature is not organized only around media formats or journalistic genres. Instead, it is organized around public problems in which news narratives play an interpretive role.

This finding can be compared with Okazaki and Takahashi (2025), where BERTopic is used to identify business-unit trends from news articles. Their results show that BERTopic can detect timely and domain-specific patterns in large collections of news data. The present study confirms the usefulness of BERTopic for identifying structured patterns in news-related corpora, but extends its use to a different level of analysis. Rather than detecting events or corporate trends in news coverage, this thesis identifies how the scholarly literature itself organizes research on news narratives.

The difference is important. In Okazaki and Takahashi's study, the topics are evidence of external business trends. In this thesis, the topics are evidence of how academic knowledge about news media narratives is structured. This makes the contribution more conceptual than predictive. The model helps reveal which public issues have become central to narrative-oriented media scholarship and which areas remain more specialized or peripheral.

The public health and crisis-related topics also connect with the review by Warsito et al. (2025), which reports that BERT-based and BERTopic-based approaches have been widely used in contexts such as COVID-19 and social media analysis. However, the role of public health in this thesis is different. It is not treated only as a computational application domain, but as one thematic area within a broader field of media narrative research. This difference suggests that health crises have become important not only for NLP applications, but also for understanding how journalism and media scholarship examine uncertainty, risk, public trust, and collective interpretation.

5.0.4 Embedding Model Selection and the Importance of Task-Specific Evaluation

The selection was based on the evaluation criteria defined in the methodology, rather than on a single performance indicator. This point is especially relevant when compared with Badieah et al. (2024). Their study reports strong performance using `all-mpnet-base-v2`, particularly when titles and abstracts are combined for publication venue recommendation. At first glance, this might appear to conflict with the present study, where `all-MiniLM-L6-v2` was selected. However, the difference is methodologically reasonable. Badieah et al. evaluate BERTopic in a recommendation setting, where the objective is to improve venue prediction. This thesis evaluates BERTopic in an interpretive setting, where the goal is to obtain topics that are coherent, diverse, not excessively fragmented, and meaningful when inspected through representative abstracts.

Therefore, the comparison does not imply that one embedding model is universally better than the other. Rather, it shows that embedding selection depends on the task, corpus, input type, and evaluation criteria. This is consistent with Warsito et al. (2025), who emphasize that transformer-based topic modeling offers advantages but also requires attention to computational feasibility, domain adaptation, and evaluation consistency. In this thesis, `all-MiniLM-L6-v2` was not selected because it is the strongest model in general, but because it provided the most suitable balance for the specific abstract-level corpus and the interpretive goals of the study.

5.0.5 BERTopic as an Interpretive Scaffold Rather Than an Automatic Answer

The role of BERTopic in this thesis is also comparable to the position developed by Kaur and Wallace (2025). Their study evaluates BERTopic against LDA and NMF for qualitative analysis of social media data and finds that researchers often prefer BERTopic because it produces more coherent, detailed, and useful clusters. However, they also emphasize that topic modeling should support qualitative interpretation rather than replace it.

This point is central to the present study. The BERTopic model identifies semantic groupings of abstracts, but the meaning of those groupings depends on the interpretation of representative terms, representative abstracts, and the broader research question. For this reason, the topics were not treated as final categories produced automatically by the algorithm. They were reviewed as thematic structures that required interpretation, validation, and careful labeling.

This approach differs from applications where BERTopic is used mainly as a classification or retrieval tool. In this thesis, BERTopic functions as an interpretive scaffold: it reduces the complexity of the corpus, exposes latent semantic groupings, and provides evidence for human analysis. This is precisely where the study contributes methodologically. It shows how BERTopic can be used in a thesis-level research design that combines quantitative diagnostics with qualitative validation.

5.0.6 Outliers as a Sign of Thematic Heterogeneity

In this corpus, the outlier class is interpretable as a consequence of disciplinary and thematic heterogeneity. This finding relates to the limitations described by Warsito et al. (2025), who note that topic modeling with transformer-based embeddings may be affected by domain variation, noisy text, and heterogeneous corpora. Although the present corpus is not composed of social media posts, it is heterogeneous in another sense: it brings together journalism studies, public health communication, environmental communication, political discourse, computational methods, and media representation. This breadth makes some abstracts difficult to assign to a stable topic without forcing weak semantic associations.

The treatment of outliers therefore strengthens rather than weakens the analysis. Instead of assigning every document to a topic, the model preserves a distinction between dense thematic structures and peripheral or highly specialized texts. This decision is consistent with the interpretive caution recommended by Kaur and Wallace (2025), since the researcher should avoid presenting all model outputs as equally stable or equally meaningful.

5.0.7 LLM-Assisted Topic Labeling and the Question of Researcher Control

The use of OpenAI-assisted labeling introduces an additional methodological layer. The LLM was not used to generate the clusters or assign documents to topics. Those steps were completed by BERTopic through embeddings, UMAP, HDBSCAN, and c-TF-IDF. The LLM was used only after the statistical topic structure had been created, as a support tool for producing concise academic labels based on representative terms and representative abstracts.

This distinction is important when compared with Kaur and Wallace (2025). Their study highlights that qualitative researchers value topic modeling when it helps organize large datasets, but they remain concerned about losing analytical control. The labeling strategy used in this thesis responds to that concern by keeping the researcher in the validation loop. The LLM proposes labels, but the labels remain dependent on the evidence supplied to the model and on subsequent researcher review.

The procedure also connects with Warsito et al. (2025), who identify LLM integration as a promising direction for topic modeling workflows. However, the present study treats this integration cautiously. LLM-assisted labels improve readability and academic consistency, but they also introduce reproducibility issues related to API access, model versioning, and prompt sensitivity. For that reason, the labeling prompt, representative evidence, and generated outputs were documented as part of the workflow. This documentation is a methodological contribution because it makes the interpretive step auditable rather than implicit.

5.0.8 Position of This Study in Relation to Previous Work

Taken as a whole, the comparison with previous studies shows that this thesis occupies a distinct position. Okazaki and Takahashi (2025) demonstrate the value of

BERTopic for extracting trends from news data. Badieah et al. (2024) show that embedding and input choices affect BERTopic performance in a recommendation task. Warsito et al. (2025) synthesize the growing role of embeddings and transformer-based topic modeling in short-text analysis. Kaur and Wallace (2025) evaluate BERTopic as a tool that can support qualitative interpretation.

This thesis draws from all these directions but does not replicate any of them. Its contribution lies in applying BERTopic to the scientific literature on news media narratives, integrating bibliometric analysis with abstract-level topic modeling, comparing model configurations, reviewing outliers, and using LLM-assisted labeling as a documented interpretive support step. In doing so, it shows that BERTopic can be used not only to analyze news texts, social media posts, or publication venues, but also to map the intellectual structure of a research field concerned with narrative meaning in news media.

Chapter 6

Conclusions

This study aimed to model and analyze narrative structures in scientific publications related to news media using BERTopic, with the purpose of identifying latent thematic patterns within the corpus. In response to the research question, the results show that the literature is organized around several recurring narrative domains, including environmental communication, trust and objectivity in news media, geopolitical narratives, migration, public health, gender-based violence, racialized violence, religious representation, and computational approaches to media analysis.

The BERTopic model identified 25 interpretable topics from the abstract corpus. These topics suggest that narrative analysis in news media research is not limited to journalism as a professional practice, but operates as a broader analytical framework for examining how public meaning is constructed around risk, conflict, identity, trust, technology, and social representation. In this sense, storytelling and narrative appear as mechanisms through which news media organize complex social, political, environmental, and technological issues.

Methodologically, the study incorporated a model comparison stage to support the selection of the final BERTopic configuration. The selected model, `all-MiniLM-L6-v2_g1`, offered the most suitable balance between topic coherence, topic diversity, outlier ratio, silhouette score, and qualitative interpretability. This comparison strengthened the reproducibility of the analysis by making the choice of hyperparameters explicit rather than relying on a single arbitrary model configuration.

In addition, OpenAI-assisted labeling was used as a complementary tool to improve the readability of the topic labels. This procedure did not modify the clusters produced by BERTopic; rather, it supported the translation of c-TF-IDF terms and representative abstracts into concise academic labels. The use of this labeling stage helped make the results more interpretable while preserving the statistical structure generated by the model.

Overall, the findings indicate that BERTopic is a useful and flexible framework for studying narrative patterns in large-scale academic corpora. The combination of embedding-based topic modeling, model diagnostics, representative terms, and assisted topic labeling provides a reproducible approach for connecting computational text analysis with interpretive questions in media and journalism studies.

6.1 Future Work

Future research could extend this study in several directions. First, applying BERTopic to full-text articles would make it possible to analyze narrative structures with greater depth and to compare whether abstracts and full texts produce similar thematic configurations. Second, temporal extensions of BERTopic could be used to examine how narrative patterns evolve across time, particularly in relation to major political, technological, environmental, or public health events.

Third, future studies could compare BERTopic with other topic modeling approaches, including LDA, dynamic topic models, and large language model-assisted topic labeling. Such comparisons would help clarify the advantages and limitations of embedding-based topic modeling for narrative analysis. Finally, expanding the corpus to include other databases, languages, or media domains would allow researchers to assess whether the thematic patterns identified in this study are specific to Scopus-indexed scholarship or reflect broader trends in research on storytelling, narratives, and news media.

Suggested Readings for Further Research

Researchers interested in extending this work may follow four complementary methodological directions.

Bibliometric analysis. This line is useful for studying how a scientific field is structured, who its main contributors are, which journals and countries lead production, and how collaboration networks are formed. For this purpose, Aria and Cuccurullo (2017) and Donthu, Kumar, Mukherjee, and Lim (2021) provide the main methodological foundations. Applied examples can be found in the bibliometric studies by Llinás and collaborators on p-adic theory and Marshall–Olkin models (Llinás, Gutiérrez, Torresblanca, De La Hoz, & Llinás, 2025; Llinás, Llinás, López, & Núñez, 2026).

Latent Dirichlet Allocation. LDA is relevant for researchers who want to compare embedding-based models with classical probabilistic topic modeling. It provides a useful baseline for understanding how topics can be inferred from word distributions, even though it does not capture semantic context as directly as BERTopic (Blei et al., 2003). The Marshall–Olkin study by Llinás et al. (2026) is a clear applied reference.

BERTopic methodology. BERTopic should be considered when the objective is to identify topics using semantic similarity rather than simple word frequency. Its combination of embeddings, dimensionality reduction, clustering, and c-TF-IDF makes it suitable for large and heterogeneous text corpora (Grootendorst, 2022). The study by Raman, Pattnaik, Hughes, and Nedungadi (2024) offers a useful example of BERTopic applied within a scientometric research design. [1]

BERT and semantic representations. BERT-based models are important for understanding why modern topic modeling can move beyond bag-of-words assumptions. These models represent documents in continuous semantic spaces, allowing texts with similar meanings to be grouped even when they use different vocabulary (Devlin et al., 2019; Reimers & Gurevych, 2019; Vaswani et al., 2017).

LLM-assisted topic labeling. BERTopic provides examples of this workflow with OpenAI models and other generative models.[2]

[1] See the official BERTopic documentation: BERTopic documentation.

[2] See the BERTopic guide on LLM-based topic representation: BERTopic LLM representation docs.

Researchers who want a cloud-based and API-driven workflow may consult the OpenAI model documentation, particularly for lightweight labeling tasks using models such as `gpt-4o-mini`.[3] Alternatively, researchers interested in a local and API-free workflow may explore Meta's Llama 3.1 models, especially the instruction-tuned variants, which can be used through local inference tools such as Ollama or llama.cpp.[4]

[3]See the OpenAI API model documentation: OpenAI GPT-4o mini model docs.
[4]See Meta's official Llama 3.1 documentation: Llama 3.1 model card and prompt formats.

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